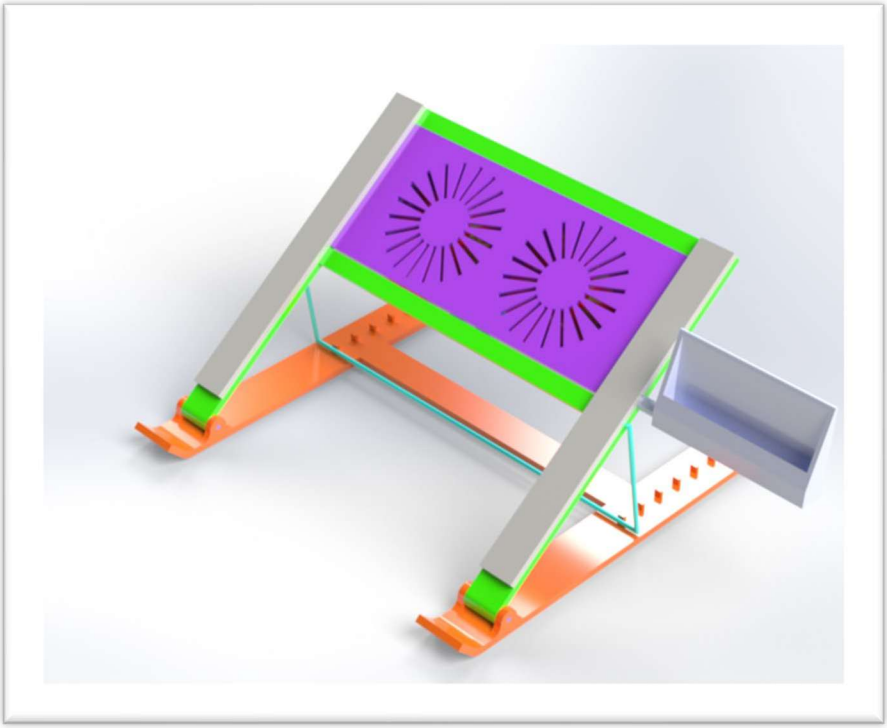
NOVEL DESIGN OF A LAPTOP STAND WITH MODULAR COOLING PAD



Design for Manufacturability

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2nd December 2024

EXECUTIVE SUMMARY

A novel laptop stand with a modular cooling pad was developed to create an ergonomic, efficient, and user-friendly product while optimizing its assembly and manufacturing processes. This project started with a benchmarking study of existing laptop stands and cooling pads to identify strengths and shortcomings in current market offerings. Using these insights, we designed a new product from scratch, modeled in SolidWorks, to refine its structure and functionality.

The design features a modular cooling pad detachable from the laptop stand, providing flexibility for users who may not always require active cooling. The stand was engineered to support various laptop sizes and provide optimal ventilation. The modular cooling pad includes strategically placed fans for efficient heat dissipation. It now features a repositioned USB hub, moving from the bottom to the top, for easier access and better cable management. After finalizing the design, we conducted a **Design for Assembly (DFA)** analysis to streamline the assembly process and identify opportunities for simplifying the product. This analysis resulted in significant improvements, including transforming the stand's frame into six easily assembled parts and reducing it to two main components for simplified production. The modular cooling pad was designed for quick attachment and detachment, enhancing user convenience while minimizing assembly time.

A thorough material selection analysis ensured the design was cost-effective and environmentally sustainable. Each part was evaluated for strength, weight, cost, and environmental impact. Lightweight plastics were chosen for the frame due to their durability, moldability, and ability to reduce overall product weight. For the cooling pad, a combination of lightweight aluminum for the base and plastic for the housing was selected to balance performance and cost. A comprehensive **manufacturing process selection analysis** followed, evaluating each component's most suitable production techniques. Injection molding was identified as the optimal method for the plastic components, while die-casting was recommended for the aluminum parts to ensure precision and quality. Additionally, rotational molding was considered to reduce material waste and production costs for larger, intricate components.

The final design incorporated two fundamental changes for improved usability and manufacturability: (1) the frame's transformation into two primary components for easier assembly and transport, (2) the redesigned cooling pad to make it smaller, (3) the USB hub being integrated into the cooling pad housing and (4) the redesign of the riser stands and phone stand holder. These changes streamlined production and assembly processes while improving the product's user experience.

This novel laptop stand has a modular cooling pad and represents a well-rounded solution that combines innovative design with cost-effective and sustainable manufacturing. By addressing assembly, material optimization, and usability, this product meets the needs of modern users while maintaining high standards for quality and efficiency.

Table of Contents

EXECUTIVE SUMMARY ii
TABLE OF FIGURES iv
TABLE OF TABLES v
PROJECT DESCRIPTION AND OBJECTIVES 1
GAANT CHART 2
BLACK BOX DIAGRAM 3
GLASS BOX DIAGRAM 3
FISHBONE DIAGRAM 4
PATENT SEARCH 4
DESIGN DOCUMENTATION 7
MISCELLANEOUS PARTS TECHNICAL SPECIFICATIONS 11
DFA ANALYSIS 12
FORMULAS USED IN DFA ANALYSIS 13
MATERIAL ANALYSIS 14
ECONOMIC ANALYSIS 22
FORMULAS 23
MANUFACTURING PROCESS ANALYSIS 28
DESIGN CHANGES 32
NEW DFA ANALYSIS 37
NEW MATERIAL AND MANUFACTURING PROCESS ANALYSIS 37
NEW ECONOMIC ANALYSIS 40
FORMULAS 40
BREAK-EVEN ANALYSIS 43
ETHICAL CONSIDERATION AND SAFETY 45
CONCLUSIONS 46
ACKNOWLEDGEMENTS 46
REFERENCES 47
APPENDICES 47

TABLE OF FIGURES

Figure 1: GAANT Chart 2
Figure 2: Black Box Diagram 3
Figure 3: Glass Box Diagram 3
Figure 4: Fishbone Diagram 4
Figure 5: Exploded View of the Cooling Pad 5
Figure 6: Cooling Fan Position 6
Figure 7: Alternate Design Fan Position 6
Figure 8: Initial Model with Cooling Pad Assembly Render 7
Figure 9: Laptop Stand Initial Model Render 7
Figure 10: Initial Design of Cooling Pad 8
Figure 11: Exploded View of the Assembly 8
Figure 12: Cooling Pad Exploded View 9
Figure 13: Laptop Stand Exploded View 9
Figure 14: USB Module on Cooling Pad 10
Figure 15: Initial DFA Analysis 12
Figure 16: Ashby Chart-ABS 14
Figure 17: Ashby Chart-Aluminum 15
Figure 18: Ashby Chart-Stainless Steel 15
Figure 19: Ashby Chart-Rubber 16
Figure 20: Ashby Chart-EVA Foam 16
Figure 21: Initial Economic Analysis 24
Figure 22: Initial Economic Analysis Cont'd 24
Figure 23: Initial Economic Analysis Cont'd 25
Figure 24: Initial Economic Analysis Cont'd 26
Figure 25: Influence of Materials on Process Selection 28
Figure 26: Manufacturing Process V/s Economic Batch Size 29
Figure 27: Modular Cooling Pad Before 32
Figure 28: Modular Cooling Pad After 32
Figure 29: Mobile Stand before 33
Figure 30: Mobile Stand After 33
Figure 31: Crossbar Support Before 34
Figure 32:Crossbar Support After 34
Figure 33: Rise Stand Before 35
Figure 34: Rise Stand After 35
Figure 35: USB Module Hub Before 36
Figure 36: USB Module After 36
Figure 37: Final DFA Analysis 37

Figure 38: Final Economic Analysis 41
Figure 39: Final Economic Analysis-Cont'd 42
Figure 40: Final Economic Analysis-Cont'd 43

TABLE OF TABLES

Table 1: Preliminary Bill of Materials 10
Table 2: Ashby Chart Ratio-Cooling Pad Fan Housing 17
Table 3: Decision Matrix- Cooling Pad Fan Housing 17
Table 4: Pugh Matrix-Cooling Pad Fan Housing 17
Table 5: Ashby Chart Ratio - Base Stand 18
Table 6: Decision Matrix-Base Stand 18
Table 7: Pugh Matrix- Base Stand 19
Table 8: Ashby Chart Ratio- Rubber Grips 19
Table 9: Decision Matrix-Rubber Grips 19
Table 10: Pugh Matrix- Rubber Grip 20
Table 11: Ashby Chart Ratio- Hinge Rod 20
Table 12: Decision Matrix-Hinge Rod 20
Table 13: Pugh Matrix- Hinge Rod 21
Table 14: Ashby Chart- Memory Foam 21
Table 15: Decision Matrix-Memory Foam 21
Table 16: Pugh Matrix- Memory Foam 22
Table 17: OME and Breakeven Analysis 27
Table 18: Break-even Analysis for Product 27
Table 19: Final Design OME and Break-Even Analysis 44
Table 20: New Breakeven Analysis for Product 44

PROJECT DESCRIPTION AND OBJECTIVES

The design and development of a novel laptop stand with a modular cooling pad were undertaken to address user needs for ergonomics, functionality, and portability. The laptop stand features a detachable cooling pad with strategically placed fans for efficient heat dissipation alongside a repositioned USB hub for enhanced accessibility. The initial design included 21 unique components, primarily plastic and lightweight aluminum parts, presenting opportunities for simplification and cost reduction. The objectives of this project were to design a unique product from scratch, perform detailed CAD modeling, and evaluate its assembly, material selection, and manufacturing processes to optimize cost and efficiency.

The stand's initial design had 35 parts, and our objective was to reduce the number of parts and redesign the stand's cooling pad with as few as possible.

Objective Technical Specifications:

1. **Component Modeling:**
    - Dimensional tolerance between ±0.005 to ±0.01 inch for all parts in SolidWorks models.
2. **Material Selection:**
    - 35 parts total, 21 unique parts which, when split according to the material, are 8 in ABS, 8 in Aluminum, 2 in steel, and 1 each in Ferromagnet (Iron), EVA, and Rubber.
    - Materials must meet ASTM standards for durability and performance.
3. **Manufacturing Process:**
    - Injection molding for plastic parts.
    - Forging for metal parts, with tolerances of ±0.05 inch.
    - Evaluate rotational molding, die casting, and laser cutting as alternatives.
4. **Part Reduction and DFA:**
    - Reduce part count by 10-15%.
    - Revamping the entire design by merging the base plate, angle stand, and riser stand. Incorporating the Modular Cooling Pad within the revamped design.
5. **Assembly Efficiency:**
    - Reduce assembly time by 20%.
    - Consolidate into three parts for the laptop stand and 2 for the modular cooling pad, improving assembly time considerably and simplifying use.

6. **Design Modifications:**
    - Secure single wire holder with a tolerance of ±0.05 inch.
    - Merge base and base plate.
    - Attach the sensor to hatch with a tolerance of ±0.05 inch.
    - Integrate PCB projections with a tolerance of ±0.01 inch.
7. **Cost Efficiency:**
    - Reduce material and production costs by 10-15%.
    - Achieve a 20% reduction in assembly time while maintaining product quality.

GANTT CHART

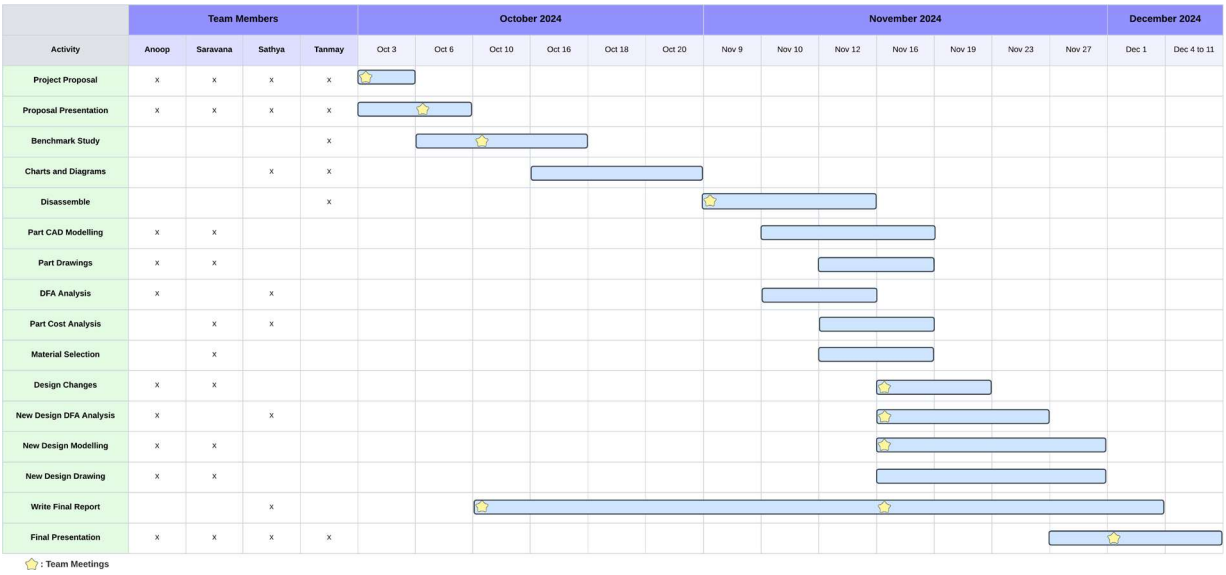


Figure 1: GANTT Chart

The GANTT chart in Figure 1 outlines the project tasks, how they were split among the group, and the deadlines for each task. The end goal was to complete the project. The milestones were various assignments, with the project report and presentation being the final milestones.

BLACK BOX DIAGRAM

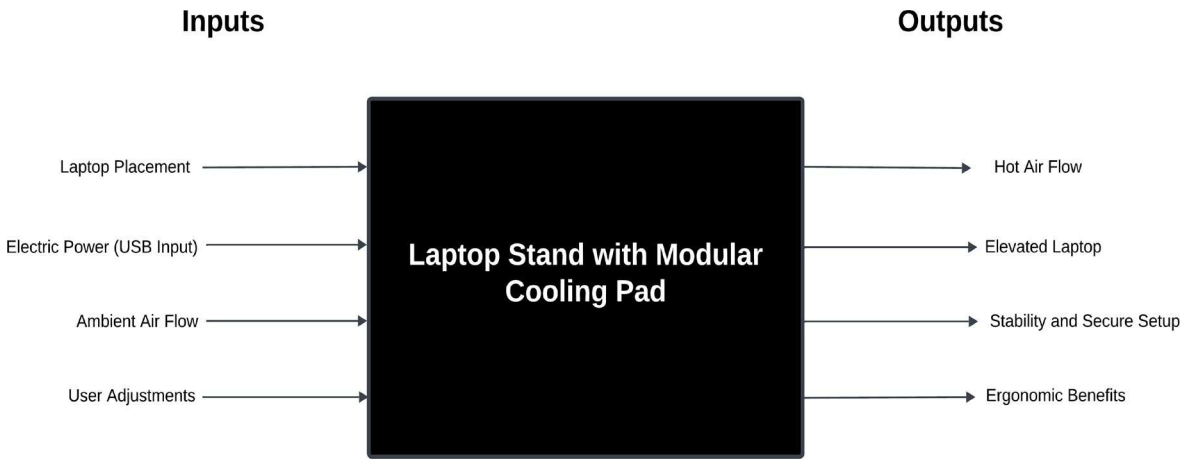


Figure 2: Black Box Diagram

The black box diagram in Figure 2 illustrates the initial conceptualization of our novel laptop stand with a modular cooling pad. Developed at the beginning of the project, this diagram was essential for evaluating the product's core functions without delving into the complexities of its internal subsystems. The primary inputs include electrical power supplied via USB to the cooling pad and placing a laptop onto the stand. User adjustments to the stand's height and angle are also considered inputs. The outputs are effective heat dissipation, improved ergonomic positioning for the user, enhanced accessibility due to the repositioned USB hub, and overall laptop stability during use. By focusing solely on inputs and outputs, the black box diagram allowed us to understand and optimize the product's functionality from a user-centric perspective before proceeding to detailed design and engineering.

GLASS BOX DIAGRAM

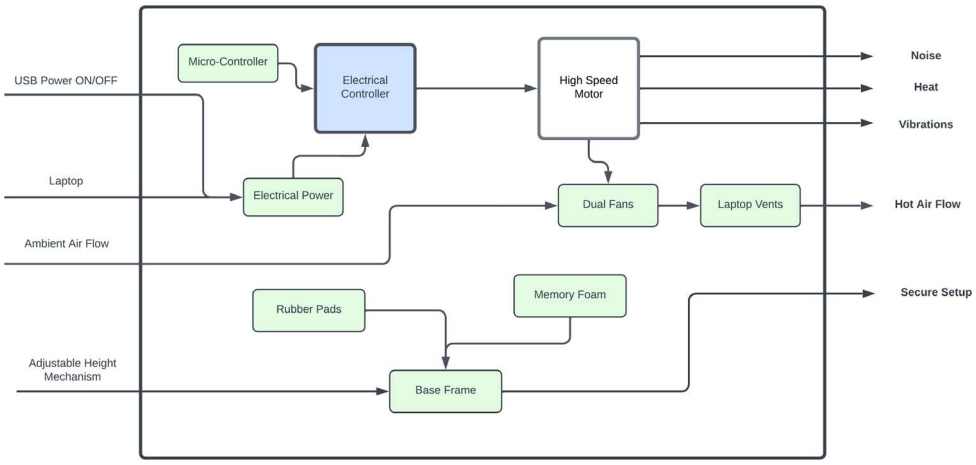


Figure 3: Glass Box Diagram

The glass box diagram in Figure 3 reveals the internal components and assemblies of the laptop stand with a modular cooling pad. This diagram represents a detailed breakdown of the internal systems, highlighting their functions, positions, and interconnections. The electrical controller, managed by a microcontroller, regulates the high-speed motor driving the dual fans, which work in tandem with the laptop vents to produce efficient heat dissipation and airflow. The base frame, equipped with rubber pads and memory foam, ensures stability and a secure setup while supporting user adjustments to height and angle. Key by-products of the system include noise, heat, and vibrations, which result from motor and fan operations. The primary outputs, however, focus on functional benefits such as hot air flow and a stable, ergonomic setup for the laptop. This diagram effectively maps the flow of inputs, control mechanisms, and outputs, providing a comprehensive understanding of the product's internal functionality.

FISHBONE DIAGRAM

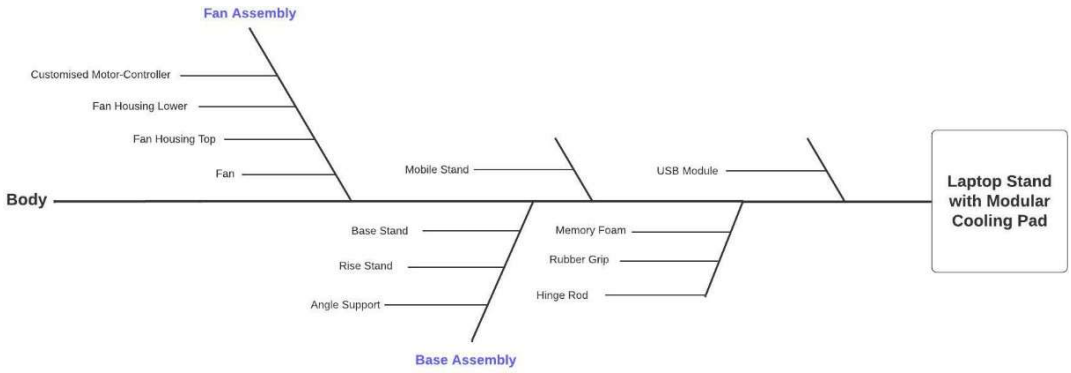


Figure 4: Fishbone Diagram

The diagram in **Figure 4** illustrates the major assemblies of the laptop stand with a modular cooling pad, divided into the **Fan Assembly** and the **Base Assembly**. The **Fan Assembly** includes components such as the customized motor controller, fan housing (top and lower), and the fan itself, which provides efficient airflow and cooling functionality. The **Base Assembly** comprises the base stand, rise stand, angle support, memory foam, rubber grips, and hinge rods, ensuring ergonomic adjustments, stability, and user comfort. These assemblies are integrated with a USB module and optional mobile stand, highlighting the modular and functional design of the product.

PATENT SEARCH

The patent, titled "Laptop Cooling Pad with Heat Dissipating Fan Adjustable in Position," **US20130114205A1**, describes a laptop cooling pad designed to improve heat dissipation. This cooling pad's unique feature is its adjustable fan system, which allows users to position one or more fans closer to heat sources specific to different laptop designs, enhancing cooling efficiency.

Key Features:

- **Adjustable Fan Positioning:** The cooling pad includes a ventilation portion with grooves that allow the fan(s) to slide horizontally (transversely) and, in some embodiments, vertically. This flexibility enables users to move the fans to directly target areas where laptops generate the most heat, like the CPU or GPU.
- **Pushing Nodes and Resilient Retainers:** Each fan is mounted on a support with pushing nodes that fit into the grooves. Users can make these nodes to adjust the fan's position manually. Resilient retainers on the fan support secure the fan in place when positioned, preventing it from moving unintentionally.
- **Power Source:** The fans are powered by connecting to the laptop's USB port, ensuring convenience and compatibility.
- **Ventilation Design:** The cooling pad's bottom and cover plates have ventilation frames that align with the fan's movement path, supporting airflow around the laptop.

This innovation addresses the limitations of traditional cooling pads with fixed fans by allowing personalized fan placement, making it adaptable to various laptop models and heat configurations.

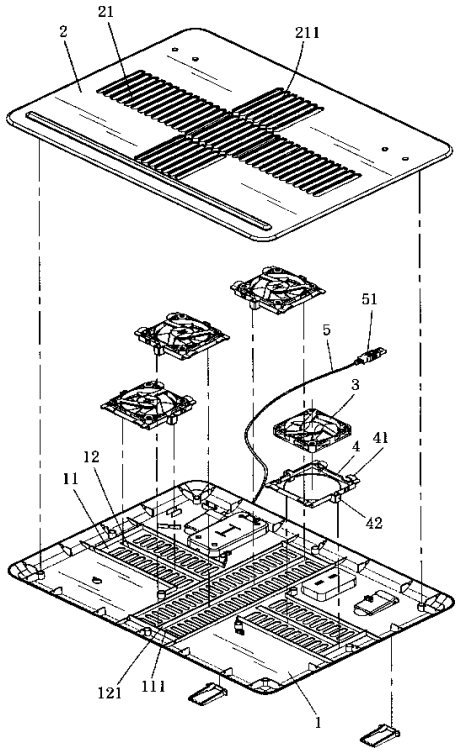


Figure 5: Exploded View of the Cooling Pad

Figures 5,6 and 7 show the two different designs for a laptop stand with a cooling pad. We took inspiration from this design and designed a simpler laptop stand with a modular cooling pad that could be detached when not needed.

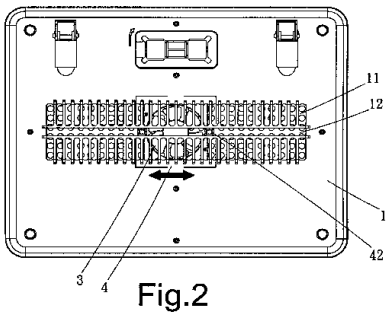


Fig.2

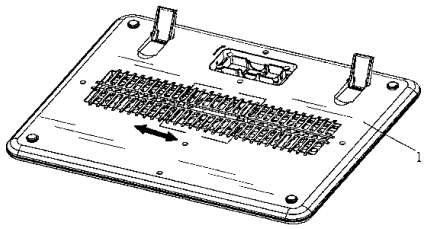


Fig.3

Figure 6: Cooling Fan Position

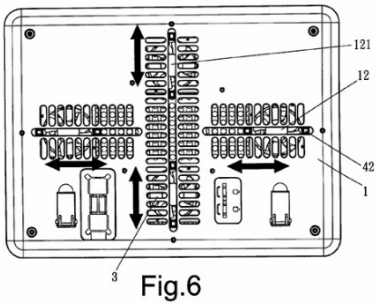


Fig.6

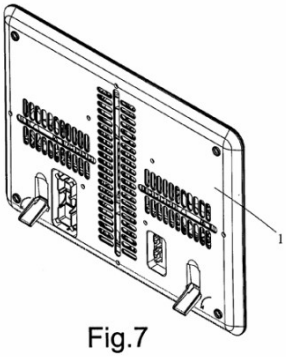


Fig.7

Figure 7: Alternate Design Fan Position

DESIGN DOCUMENTATION

The figures below are the initial model rendering.

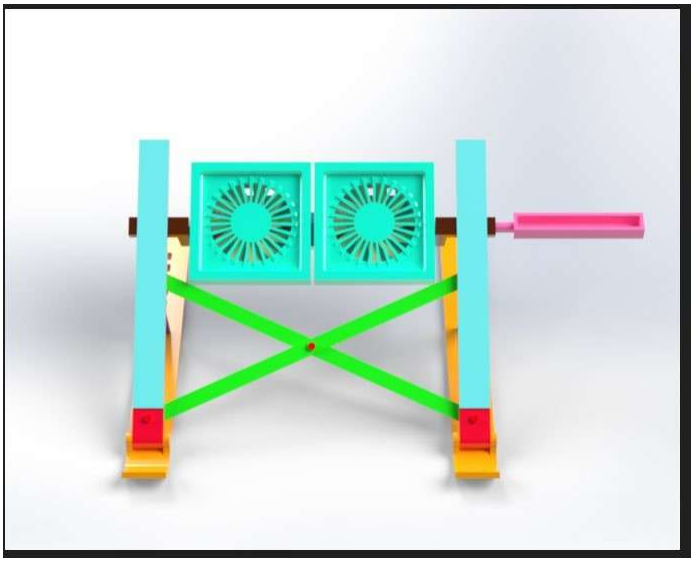


Figure 8: Initial Model with Cooling Pad Assembly Render

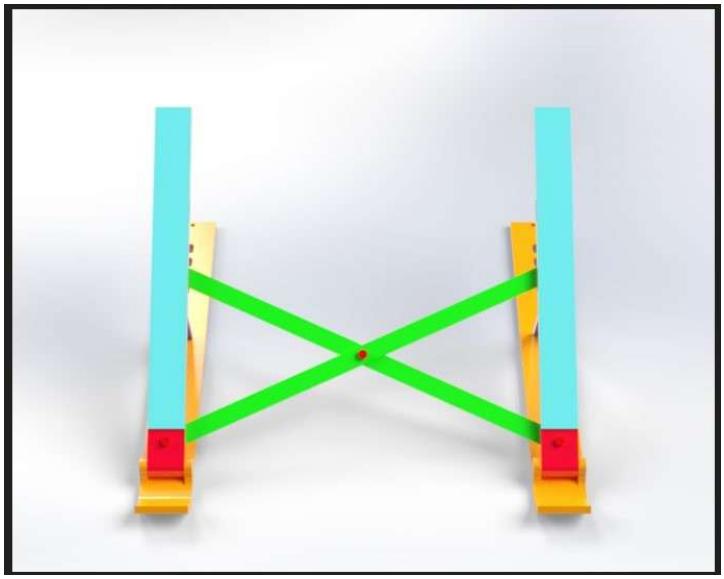


Figure 9: Laptop Stand Initial Model Render

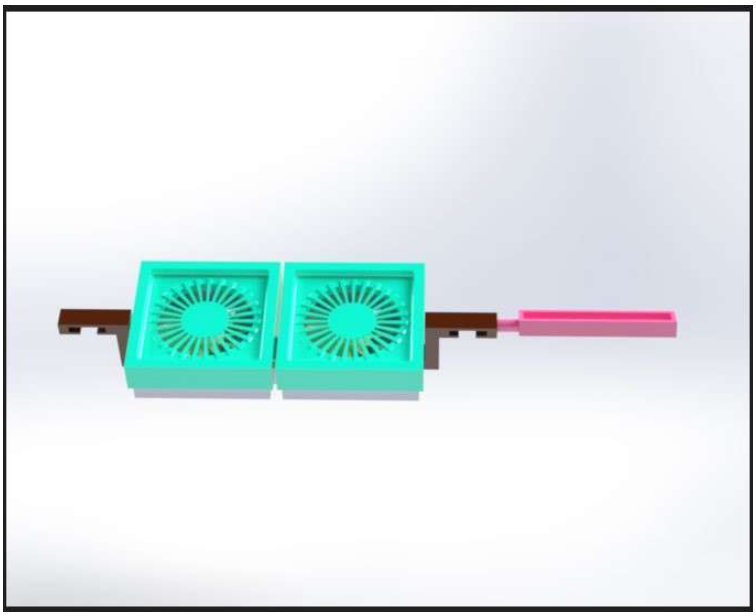


Figure 10: Initial Design of Cooling Pad

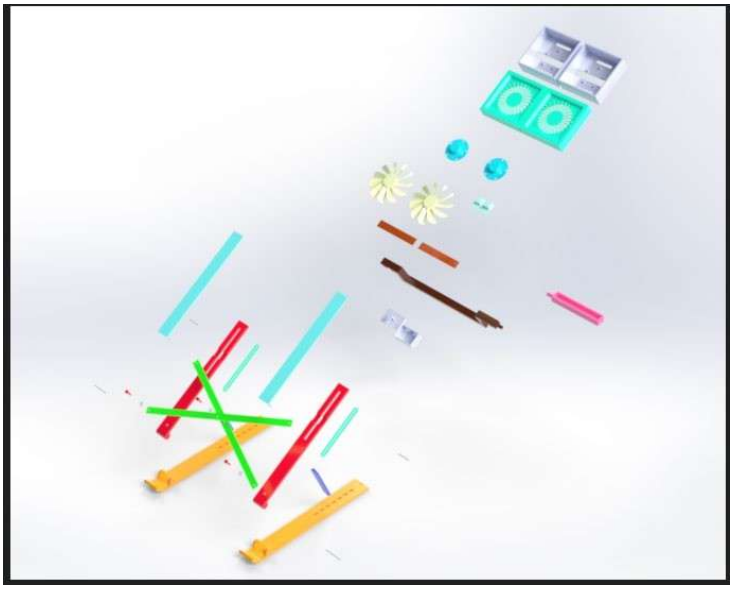


Figure 11: Exploded View of the Assembly

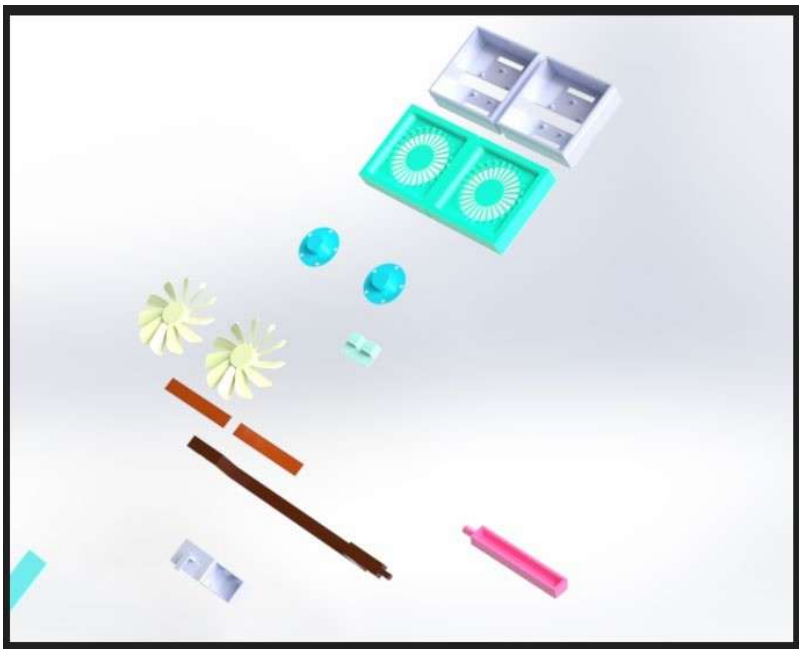


Figure 12: Cooling Pad Exploded View

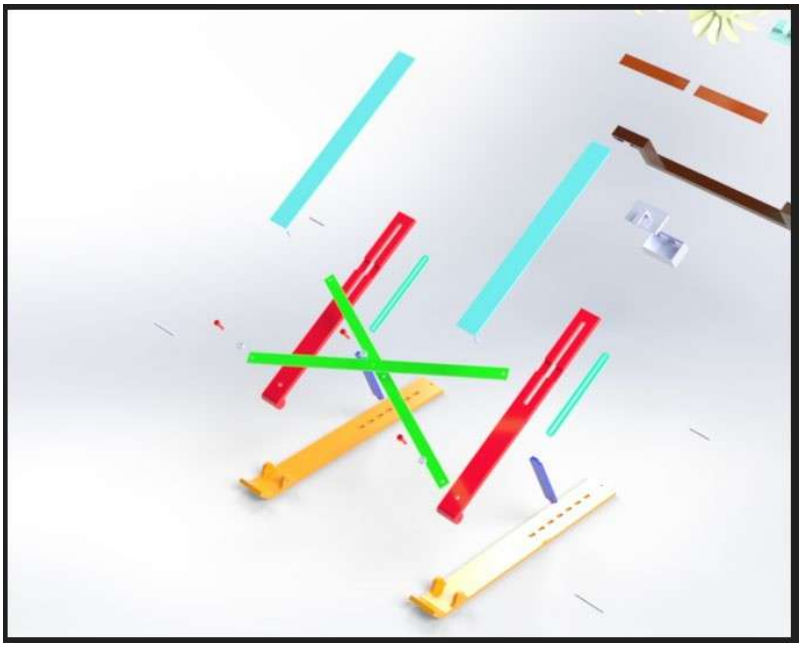


Figure 13: Laptop Stand Exploded View

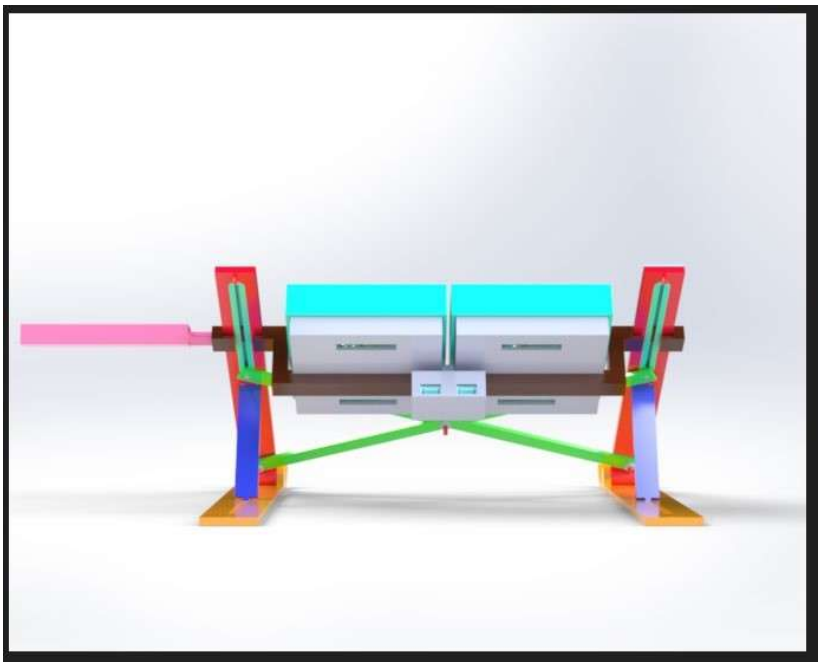


Figure 14: USB Module on Cooling Pad

The table below shows the preliminary bill of materials created after the initial design and indicates whether the part is off the shelf.

Table 1: Preliminary Bill of Materials

SL.NO	Part Number	Description	Material	Quantity	OTS
1.	LS01-01	Fan	ABS	2	YES
2.	LS01-02	Fan Housing-Top	ABS	2	-
3.	LS01-03	Fan Housing-Bottom	ABS	2	-
4.	LS01-04	Customized Motor-Controller	ABS	1	YES
5.	LS01-05	Fan Housing Carrier	Aluminum	1	-
6.	LS01-06	Magnetic Strip	Ferromagnet	2	-
7.	LS02-01	Base Stand-Left	Aluminum	1	-
8.	LS02-02	Base Stand-Right	Aluminum	1	-
9.	LS02-03	Crossbar Support	Aluminum	2	-
10.	LS02-04	Rise Stand-Right	Aluminum	1	-
11.	LS02-05	Rise Stand-Left	Aluminum	1	-
12.	LS02-06	Angle Support-Left	Aluminum	1	-
13.	LS02-07	Angle Support-Right	Aluminum	1	-
14.	LS03	Phone Stand Holder	ABS	1	-
15.	LS04-01	Hinge rod	Steel	4	YES
16.	LS04-02	Rubber grip	Rubber	4	YES
17.	LS04-03	Memory foam	EVA	2	YES
18.	LS04-04	Nut and Bolt	Steel	3	YES
19.	LS05-01	USB Hub	ABS	1	YES
20.	LS05-02	USB Hub Holder	ABS	1	-
21.	LS05-03	USB Top Cap	ABS	1	-

MISCELLANEOUS PARTS TECHNICAL SPECIFICATIONS

The cooling pad has two fan housings roughly 3.5 inches in diameter.

Sunon Fans BLDC

- **Dimensions:** (3.14 inches x 3.14 inches x 0.98 inches)
- **Manufacturer:** Sunon Fans
- **Voltage – Rated: 12VDC**
- **Airflow:** 41 CFM (1.15m^3/min)
- **Static Pressure:** 0.180 in H2O (44.8Pa)
- **Bearing Type:** Sleeve
- **Fan Type:** Tubeaxial
- **Features:** Auto-restart
- **Noise:** 33 dB(A) Max
- **Power Watts:** 1.44W
- **RPM:** 3200rpm
- **Operating Temperature:** 14 ~ 158°F (-10 ~ 70°C)
- **Weight:** 0.167 lb (75.75 g)
- **Current Rating (Ampere):** 0.12A
- **Voltage Range:** 4.5 ~ 13.8VDC
- **Material – Frame:** Polybutylene Terephthalate (PBT)
- **Material – Blade:** Polybutylene Terephthalate (PBT)
- **Lifetime @ Temp.:** 35000 Hrs @ 40°C (104F)
- **Product Datasheet: Link**

DFA ANALYSIS

DFA Analysis Worksheet

Assembly Name: Laptop stand with modular cooling pad **Team:** Group 6 **Date:** 19-11-2024

If the answer is Yes to any of the metrics or questions enter a 1. If the answer is No then enter 0. Each cell must have a number.

Part		Complexity		Functional Analysis / Redesign Opportunity				Error Proofing		Handling			Insertion				Secondary Operations				
Part Number	Part Name	Number of Parts (N_p)	Number of Interfaces (Ni)	Theoretical Minimum Part	Part Can Be Standardized (if not already standard)	Cost (Low/Medium/High)	Practical Minimum Part	Assemble Wrong Part/ Omit Part	Assemble Part Wrong Way Around	Tangle, Nest, or Stick Together	Flexible, Fragile, Sharp or Slippery	Pliers, Tweezers, or Magnifying Glass Needed	Difficult to Align/ Locate	Holding Down Required	Resistance to Insertion	Obstructed Access/ Visibility	Re-orient Workpiece	Screw, Drill, Twist, Rivet, Bend, or Crimp	Weld, Solder, or Glue	Paint, Lube, Heat, Apply Liquid or Gas	Test, Measure or Adjust
LS01	FAN ASSEMBLY																				
LS01-01	Fan	2	1	1	1	M	1	0	0	1	1	0	0	0	0	0	0	0	0	0	0
LS01-02	Fan Housing-Top	2	4	1	1	M	1	0	0	1	0	0	0	0	1	0	0	0	0	0	0
LS01-03	Fan Housing-Bottom	2	4	1	1	M	1	0	0	1	0	0	0	0	1	0	0	0	0	0	0
LS01-04	Customised Motor-Controller	1	2	1	1	L	1	0	0	1	0	0	0	0	0	0	0	1	0	0	1
LS01-05	Fan Housing Carrier	1	2	1	0	M	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
LS01-06	Magnetic Strip	2	4	1	1	L	1	0	0	1	0	0	0	0	0	0	0	0	1	0	0
LS02	BASE ASSEMBLY																				
LS02-01	Base Stand-Left	1	1	1	1	L	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0
LS02-02	Base Stand-Right	1	1	0	1	L	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
LS02-03	Crossbar Support	2	8	1	0	L	1	0	0	0	0	0	0	0	0	0	0	1	0	1	0
LS02-04	Rise Stand-Right	1	4	1	1	L	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
LS02-05	Rise Stand-Left	1	4	0	1	L	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0
LS02-06	Angle Support-Left	1	2	1	1	L	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0
LS02-07	Angle Support-Right	1	2	0	1	L	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
LS03	Phone Stand Holder	1	2	0	0	L	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0
LS04	OFF THE SHELF																				
LS04-01	Hinge rod	4	2	1	1	L	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0
LS04-02	Rubber grip	4	1	1	1	L	1	0	0	1	1	0	0	0	0	0	0	0	1	1	0
LS04-03	Memory foam	2	4	1	1	L	0	1	0	1	1	0	0	0	0	0	0	0	1	1	0
LS04-04	Nut and Bolt	3	15	1	1	L	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
LS05	USB ASSEMBLY																				
LS05-01	USB Hub	1	1	1	1	M	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
LS05-02	USB Hub Holder	1	1	0	0	L	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
LS05-03	USB Top Cap	1	2	1	0	L	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0
	Totals	35	67	16	16	0	14	1	0	7	3	0	0	0	6	0	0	2	3	10	1
	Design for Assembly Metrics	**48.43**		**45.7%**	←Theor. Effy. Pract. Effy.→		**40.0%**	**0.06**		**0.63**			**0.38**				**1.00**				
	Targets																				

Figure 15: Initial DFA Analysis

Figure 15 above shows a complete DFA analysis. The DFA metrics score is 48.43, which is moderately high. There are 35 parts and 67 interfaces.

Functional Analysis

The functional analysis evaluates the necessity of each part and the potential for redesign. With a current Part Count Efficiency of 45.7%, which indicates the number of parts used in the assembly process, there are clear opportunities for enhancement. The practical minimum part count is

significantly lower than the actual count of 35, suggesting that components like the rise stand supports and USB hub holder could be combined or eliminated. For instance, consolidating the frame components from six individual parts into two larger, integrated sections could significantly reduce assembly complexity and greatly improve functional efficiency.

Error Proofing

Our error-proofing score, currently at 0.86, indicates moderate alignment or assembly error risks for parts such as the fan housing bottom and USB hub holder. These parts lack sufficient features like alignment guides or keyed designs, which increases the likelihood of errors during assembly. However, the potential for improvement is significant. Features like snap-fit connectors for the USB hub or keyed slots for the frame parts could substantially improve this score and enhance assembly accuracy.

Handling

Our handling score, currently at 0.63, highlights the challenges in managing smaller, delicate parts like the magnetic strip and USB hub components. However, the potential for improvement is significant. We can significantly improve this score and reduce assembly time by simplifying the frame assembly and pre-assembling smaller parts like the USB module with its holder.

Insertion

Insertion evaluates the ease of aligning and placing components. With an Insertion Score of 0.33, there are clear areas for improvement, particularly for parts that demand precise alignment, such as the fan assembly and cooling pad attachment system. Incorporating alignment features like guide pins, tapered edges, or locating grooves for these components could streamline the insertion process, leading to faster and more reliable assembly.

Secondary Operations

Secondary operations assess tasks such as adhesive application, soldering, or finishing. The **Secondary Operations Score is 0.06**, showing minimal additional steps are required. However, parts like the rubber grips require an adhesive application, and screws are used in the USB hub holder, which could be redesigned for tool-free assembly to simplify the process further and reduce time.

FORMULAS USED IN DFA ANALYSIS

- DFA Metric= $\sqrt{Number\ of\ Parts * Number\ of\ Interfaces}$
- Theoretical Efficiency= $\frac{Theoretical\ Minimum\ Part}{Number\ of\ Parts}$
- Practical Efficiency= $\frac{Practical\ Minimum\ Part}{Number\ of\ Parts}$
- Error Proofing= $\frac{Assemble\ WronWW\ Part\ or\ OOmit\ Part + Assemble\ Part\ WronWW\ Way\ Around}{Theoretical\ Minimum\ Part}$
- Handling= $\frac{(TanWWle, Nest\ or\ Stick\ ToWWether) + (Flexible, FraWWile, Sharp\ or\ Slippery) + (Pliers, TTTeezers, or\ MaWWnifyinWW\ Glass\ Needed)}{Theoretical\ Minimum\ Part}$

- Insertion=$\frac{Difficult\ to\ AliWWn+HoldinW\ W\ DoTTn\ Required+Resistance\ to\ Insertion+OObstructed\ Access}{Theoretical\ Minimum\ Part}$
- Secondary Operation=

$$\frac{Reorient\ \boldsymbol{WW}orkpiece+(Scre\boldsymbol{SS},Drill,T\boldsymbol{SS}ist,Rivet,Bend\ or\ Crimp)+(\boldsymbol{WW}eld,Solder\ or\ GLue)+(Paint,Lube,Heat,Apply\ Li\boldsymbol{LL}uid\ or\ Gas)+(Test,Measure\ or\ Ad\boldsymbol{AA}ust)}{Theoretical\ Minimum\ Part}$$

MATERIAL ANALYSIS

We conducted a material analysis to determine the materials used in our laptop stand with cooling pad. We found that the components were evenly matched between ABS and Aluminum with few components being Steel and some off the shelf components being Rubber and EVA Foam.

A comparison analysis was done to determine if other materials could be used in the product. We used an Ashby chart to determine materials similar in strength that could be used in our product. The Ashby charts for the different materials are shown in Figures 16-20 below.

- $\frac{\boldsymbol{\sigma_f^{22/33}}}{\boldsymbol{\rho}}$
- $\frac{\boldsymbol{E^{1/22}}}{\boldsymbol{\rho}}$

E= Modulus of Elasticity, $\boldsymbol{\sigma_f}$ = *Failure Strength*, $\boldsymbol{\rho}$ = *Density*

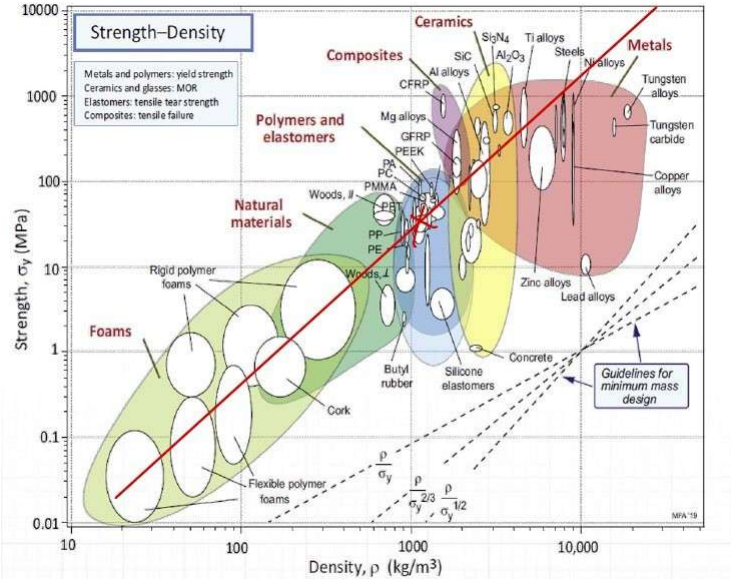


Figure 16: Ashby Chart-ABS

- $\frac{\boldsymbol{\sigma_f^{22/33}}}{\boldsymbol{\rho}}$
- $\frac{\boldsymbol{E^{1/22}}}{\boldsymbol{\rho}}$

E= Modulus of Elasticity, $\boldsymbol{\sigma_f}$ = *Failure Strength*, $\boldsymbol{\rho}$ = *Density*

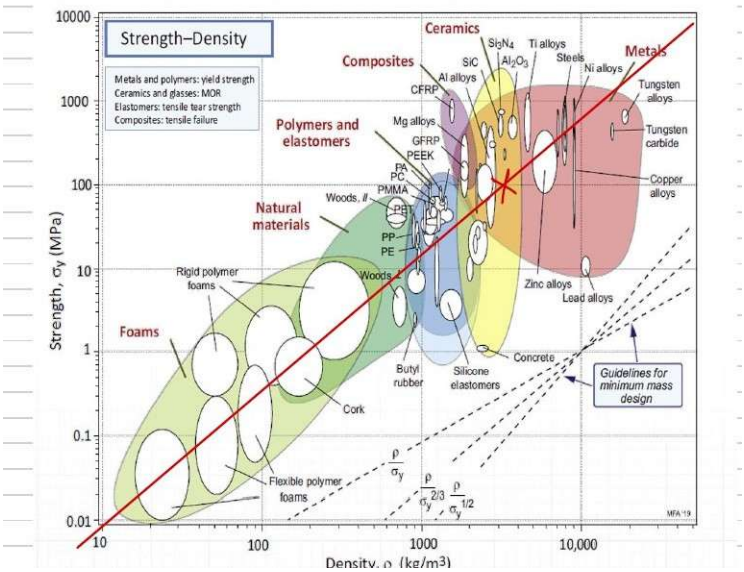


Figure 17: Ashby Chart-Aluminum

- $\frac{\sigma_f^{22/33}}{\boldsymbol{\rho}}$
- $\frac{E^{1/22}}{\boldsymbol{\rho}}$

E= Modulus of Elasticity, $\boldsymbol{\sigma_f} = Failure\ Strength, \boldsymbol{\rho} = Density$

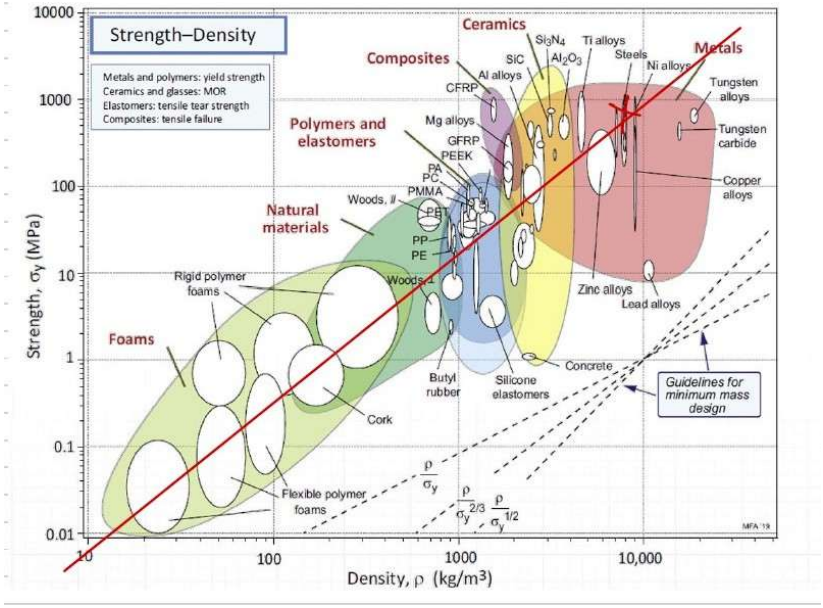


Figure 18: Ashby Chart-Stainless Steel

- $\frac{\sigma_f^{22/33}}{\boldsymbol{\rho}}$
- $\frac{E^{1/22}}{\boldsymbol{\rho}}$

E= Modulus of Elasticity, $\sigma_f = Failure\ Strength, \rho = Density$

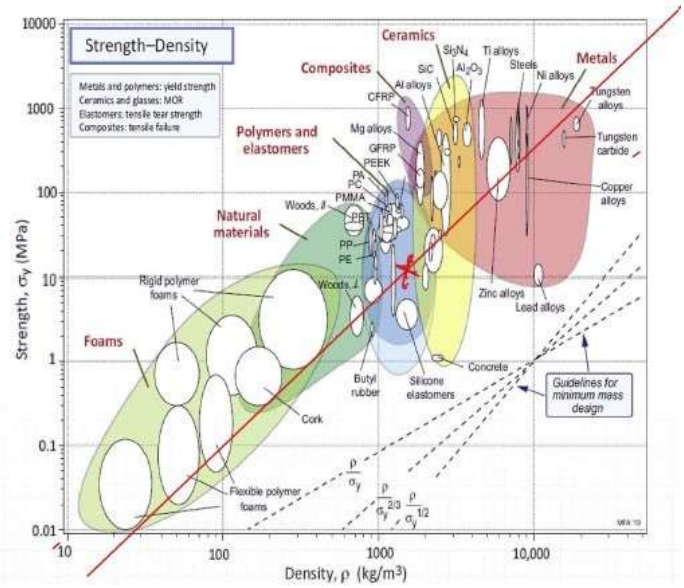


Figure 19: Ashby Chart-Rubber

- $\frac{\sigma_f^{22/33}}{\rho}$
- $\frac{E^{1/22}}{\rho}$

E= Modulus of Elasticity, $\sigma_f = Failure\ Strength, \rho = Density$

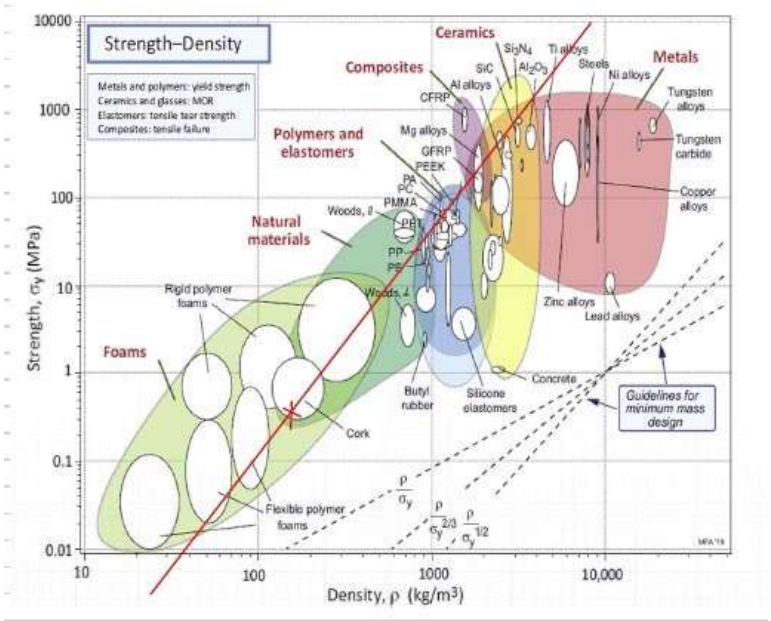


Figure 20: Ashby Chart-EVA Foam

1. **Cooling Pad Fan Housing (ABS)**

Materials Considered:

- Polypropylene (PP)
- Polycarbonate (PC)
- ABS

Ashby Chart Ratio (Strength-to-Weight Formula):

Table 2: Ashby Chart Ratio-Cooling Pad Fan Housing

Material	Strength (MPa)	Density (kg/m³)	Ashby Ratio (strength^2/3 / density)
PC/ABS	60	1100	0.099
Polycarbonate (PC)	70	1200	0.110
ABS	45	1040	0.094

Decision Matrix:

Table 3: Decision Matrix- Cooling Pad Fan Housing

Material	Strength-to-Weight	Cost	Manufacturability	Durability	Corrosion Resistance	Weighted Score
PP	6	8	8	6	5	6.6
PC	7	5	6	7	6	6.2
ABS	6	7	9	7	6	7.0

Here, the weighted average score is the average of all the criteria scores. It represents how good a material is out of 10.

Pugh Matrix:

Table 4: Pugh Matrix-Cooling Pad Fan Housing

Criteria	Baseline (ABS)	PP	PC
Strength-to-Weight	0	0	+
Cost	0	+	-
Manufacturability	0	0	-
Durability	0	-	+

Criteria	Baseline (ABS)	PP	PC
Corrosion Resistance	0	-	0

Here, + means better, - means worse, and 0 indicates the same.

Conclusion:

ABS is still the obvious choice, but PC or PP are close in terms of properties and cost. Replacing ABS is tough unless PC or PP are reduces cost to manufacture in big batches.

2. **Base Stand (Aluminum Components)**

Materials Considered:

- Aluminum
- Stainless Steel
- Titanium
- Magnesium Alloy

Material Options and Ashby Chart Ratio Calculation:

Table 5: Ashby Chart Ratio - Base Stand

Material	Strength (MPa)	Density (kg/m³)	Ashby Ratio (strength^2/3 / density)
Aluminum 6061	270	2700	0.170
Magnesium	160	1800	0.177
Titanium	1000	4500	0.222
Steel	600	7900	0.111

Decision Matrix:

Table 6: Decision Matrix-Base Stand

Material	Strength-to-Weight	Cost	Manufacturability	Durability	Corrosion Resistance	Weighted Score
Aluminum (6061)	8	6	9	8	9	7.8
Magnesium	7	7	8	7	6	7.0
Titanium	10	3	6	10	10	7.8
Steel	9	4	6	9	8	7.2

Pugh Matrix:

Table 7: Pugh Matrix- Base Stand

Criteria	Baseline (Aluminum)	Magnesium	Steel	Titanium
Strength-to-Weight	**0**	+	-	++
Cost	**0**	**0**	+	--
Manufacturability	**0**	**0**	-	-
Durability	**0**	-	+	++
Corrosion Resistance	**0**	-	**0**	++

Conclusion:

Aluminum 6061 remains the preferred choice due to its excellent strength-to-weight ratio, cost-effectiveness, and ease of manufacturability. Titanium, while offering greater strength, is cost-prohibitive for this application, and magnesium, while lightweight, poses challenges in durability.

3. **Rubber Grips (Rubber)**
   Materials Considered:

- Rubber
- Silicone Rubber
- Thermoplastic Elastomer (TPE)

Table 8: Ashby Chart Ratio- Rubber Grips

Material	Strength (MPa)	Density (kg/m³)	Ashby Ratio (strength^2/3 / density)
Rubber	20	1200	0.0029
Silicone Rubber	7	1100	0.0018
TPE	10	1000	0.0024

Decision Matrix:

Table 9: Decision Matrix-Rubber Grips

Material	Strength-to-Weight	Cost	Manufacturability	Durability	Corrosion Resistance	Weighted Score
Rubber	5	9	8	6	5	6.7
Silicone Rubber	6	7	7	8	7	7.1
TPE	7	8	9	7	6	7.4

Pugh Matrix:

Table 10: Pugh Matrix- Rubber Grip

Criteria	Baseline (Rubber)	Silicone Rubber	TPE
Strength-to-Weight	**0**	+	+
Cost	**0**	-	**0**
Manufacturability	**0**	**0**	+
Durability	**0**	+	-
Corrosion Resistance	**0**	**0**	**0**

Conclusion:

TPE seems to be better than rubber in most of the criteria but is not durable. TPE may also be costlier depending on suitable manufacturing process. The logical option is to stay with Rubber, but TPE can potentially replace Rubber.

4. **Hinge Rod (Steel Components)**

Materials Considered:

- Steel
- Stainless Steel
- HSLA (High Strength Low Alloy) Steel

Table 11: Ashby Chart Ratio- Hinge Rod

Material	Strength (MPa)	Density (kg/m³)	Ashby Ratio (strength^2/3 / density)
HSLA	500	7850	0.058
Stainless Steel	450	8000	0.054
Steel	400	7900	0.057

Decision Matrix:

Table 12: Decision Matrix-Hinge Rod

Material	Strength-to-Weight	Cost	Manufacturability	Durability	Corrosion Resistance	Weighted Score
HSLA	9	6	7	8	7	7.4
Stainless Steel	8	5	6	9	9	7.4
Steel	9	4	6	9	8	7.2

Pugh Matrix:

Table 13: Pugh Matrix- Hinge Rod

Criteria	Baseline (Steel)	Stainless Steel	HSLA Steel
Strength-to-Weight	**0**	**0**	+
Cost	**0**	**-**	**0**
Manufacturability	**0**	**-**	**0**
Durability	**0**	**0**	**-**
Corrosion Resistance	**0**	+	**-**

Conclusion:

The obvious choice is to stay with steel or replace it with Stainless steel. HSLA for this application may seem too costly and pointless.

5. **Memory Foam (EVA Foam)**
   Materials Considered:
   - EVA Foam
   - Polyurethane Foam
   - Polyethylene Foam

Table 14: Ashby Chart- Memory Foam

Material	Strength (MPa)	Density (kg/m³)	Ashby Ratio (strength^2/3 / density)
EVA Foam	0.5	300	0.0031
Polyurethane Foam	0.8	400	0.0030
Polyethylene Foam	0.6	200	0.045

Decision Matrix:

Table 15: Decision Matrix-Memory Foam

Material	Strength-to-Weight	Cost	Manufacturability	Durability	Corrosion Resistance	Weighted Score
EVA Foam	4	8	9	5	4	6.3
Polyurethane Foam	6	7	8	6	5	6.8
Polyethylene Foam	5	8	9	5	6	6.8

Pugh Matrix:

Table 16: Pugh Matrix- Memory Foam

Criteria	Baseline (EVA Foam)	Polyurethane Foam	Polyethylene Foam
Strength-to-Weight	**0**	+	+
Cost	**0**	-	**0**
Manufacturability	**0**	**0**	**0**
Durability	**0**	+	**0**
Corrosion Resistance	**0**	+	+

Conclusion:

The two alternatives are evenly matched and have the potential to replace EVA Foam. Still, cost, manufacturing process, and batch size must be considered before any material changes. This is to ensure the price is as low as possible.

ECONOMIC ANALYSIS

Laptop Stands have relatively lower manufacturing costs due to their more straightforward design, but Cooling Pads may have higher manufacturing costs due to their complexity. Due to their unique design and multiuse purpose, they are widely available in different forms with various fan configurations. So, the analysis is done based on the various costs involved.

After material and process selection is accomplished, cost analysis is done to make sound technological choices and judge the suitability of suppliers' bids. This analysis helped us determine whether our redesign made financial sense. Various cost estimators were considered to determine the possible product cost.

The cost estimators that were considered are as follows:

Order of Magnitude (OME) Estimates:

This was used to roughly guess the price of each component according to the 1:3:9 rule. Initially, material costs for components were calculated using the weight, price per pound of the material, and waste factor as follows:

Material Cost = (Waste Factor * Weight * Price per pound)

Manufacturing Cost = (3 * Material Cost) Selling Price = (9 * Material Cost)

FORMULAS

ECONOMIC ANALYSIS

- $C_m=\frac{m*c_m}{1-f}$, where **C_m= Unit cost of material**, **m= mass of part, c_m=cost of material, f=fraction of material scrap**.
- $C_L=\frac{c_{ww}}{n'}$, where **C_L= Unit Labor Cost, c_w= hourly cost of wages and benefits,** n' = ppppppppppppppppppp pprrpprr.
- $C_T=\frac{ct \cdot k}{n}$ where, **C_T= Tooling Cost, c_t= cost of making tools, n= production run, k= tooling wear factor** (**k= n**/by life of tooling raised to next highest integer).
- $C_E=(\frac{1}{n'}) \cdot (\frac{c_e}{Lt_{wwo}}) \cdot q$, **where c_e= cost of capital equipment, n'= production rate, L= Load Factor, t_{wo}= capital write-off time, q= fraction of equipment sharing between products.**
- $C_{OH}=\frac{cOOH}{n'}$, **where c_{OH}= overhead hourly rate, n'= production rate**.
- C_U**= Cm+C_L+C_T+C_E+C_{OH}**

These calculations are shown in Figures 21,22,23 and 24.

Assumptions made in analysis:

For individual components, a fixed cost of $50000 was used. This cost includes manufacturing, depreciation, and other costs. We assumed a fixed cost of $750000 for the entire assembly as those costs typically include marketing, inventory, shipping, and logistics. For capital write-off time in the sheet, we assumed operations for 45 hours each for 52 weeks, and we amortized the components over 5 and 10 years based on their complexity and size. We also assumed a total product profit of $22 for product break-even analysis. With these assumptions, we did the economic analysis calculations visible below and the break-even analysis, the latter shown in Tables 17 and 18 below.

INITIAL ECONOMIC ANALYSIS								
Formula	Cost Element	Variables	Units	LS01-01	LS01-02	LS01-03	LS01-04	LS01-05
				Fan	Fan Housing-Top	Fan Housing-Bottom	Customised Motor-Controller	Fan Housing Carrier
	Material			ABS	ABS	ABS	ABS	Aluminium
Cm=m*cm/1-f	Material cost	Cm	$/lb	1.75	1.75	1.75	1.75	1.16
	is scrap	f		0.05	0.05	0.05	0.05	0.05
	Mass of part	m	lb	0.012	0.114	0.1164	0.2	0.41258
	C_m unit cost of material			**0.022**	**0.210**	**0.214**	**0.368**	**0.504**
Cl=cw/n	Labour cost	Cw	$/h	25	25	25	25	25
	Production rate	n	units/h	2000	2000	2000	2000	2000
	C_l unit cost of labour			**0.0125**	**0.0125**	**0.0125**	**0.0125**	**0.0125**
Ct=ct*k/n	Tooling cost	Ct	$/set	50000	50000	50000	50000	50000
	Total production run	n	units	250000	250000	250000	250000	250000
	Tooling life	nt	units	75000	75000	75000	75000	75000
	Sets of tooling required	k		3.333	3.333	3.333	3.333	3.333
	C_t unit cost of tooling			**0.667**	**0.667**	**0.667**	**0.667**	**0.667**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	60000	60000	60000	60000	60000
	Capital write-off time	Five	years	23400	23400	23400	23400	23400
	Load fraction	L		1	1	1	1	1
	Load sharing fraction	q		1	1	1	1	1
	C_e unit cost of capital equipment			**0.0013**	**0.0013**	**0.0013**	**0.0013**	**0.0013**
Coh=coh/n	Factory overhead	coh	$/h	50	50	50	50	50
	Production rate	n	units/h	2000	2000	2000	2000	2000
	C_{oh} unit cost of factory overhead			**0.025**	**0.025**	**0.025**	**0.025**	**0.025**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**0.728**	**0.915**	**0.920**	**1.074**	**1.209**

Figure 21: Initial Economic Analysis

INITIAL ECONOMIC ANALYSIS								
Formula	Cost Element	Variables	Units	LS01-06	LS02-01	LS02-02	LS02-03	LS02-04
				Magnetic Strip	Base Stand-Left	Base Stand-Right	Crossbar Support	Rise Stand-Left
	Material			Ferromagnet (Iron)	Aluminium	Aluminium	Aluminium	Aluminium
Cm=m*cm/1-f	Material cost	Cm	$/lb	0.4	1.16	1.16	1.16	1.16
	is scrap	f		0.05	0.05	0.05	0.05	0.05
	Mass of part	m	lb	0.03	0.3234	0.3234	0.1176	0.02842
	C_m unit cost of material			**0.013**	**0.395**	**0.395**	**0.144**	**0.035**
Cl=cw/n	Labour cost	Cw	$/h	25	25	25	25	25
	Production rate	n	units/h	2000	2000	2000	2000	2000
	C_l unit cost of labour			**0.0125**	**0.0125**	**0.0125**	**0.0125**	**0.0125**
Ct=ct*k/n	Tooling cost	Ct	$/set	50000	50000	50000	50000	50000
	Total production run	n	units	250000	250000	250000	250000	250000
	Tooling life	nt	units	75000	75000	75000	75000	75000
	Sets of tooling required	k		3.333	3.333	3.333	3.333	3.333
	C_t unit cost of tooling			**0.667**	**0.667**	**0.667**	**0.667**	**0.667**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	60000	50000	50000	50000	50000
	Capital write-off time	Five	years	23400	23400	23400	23400	23400
	Load fraction	L		1	1	1	1	1
	Load sharing fraction	q		1	1	1	1	1
	C_e unit cost of capital equipment			**0.0013**	**0.0011**	**0.0011**	**0.0011**	**0.0011**
Coh=coh/n	Factory overhead	coh	$/h	50	50	50	50	50
	Production rate	n	units/h	2000	2000	2000	2000	2000
	C_{oh} unit cost of factory overhead			**0.025**	**0.025**	**0.025**	**0.025**	**0.025**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**0.718**	**1.100**	**1.100**	**0.849**	**0.740**

Figure 22: Initial Economic Analysis Cont'd

INITIAL ECONOMIC ANALYSIS								
Formula	Cost Element	Variables	Units	LS02-05 Rise Stand-Right	LS02-06 Angle Stand-left	LS02-07 Angle Stand-Right	LS03 Phone Stand Holder	LS04-01 Hinge Rod
	Material			Aluminium	Aluminium	Aluminium	ABS	Stainless Steel
Cm=m*cm/1-f	Material cost	Cm	$/lb	1.16	1.16	1.16	1.75	1
	is scrap	f		0.05	0.05	0.05	0.05	0.05
	Mass of part	m	lb	0.02842	0.20972	0.20972	0.0784	0.0058
	C_m unit cost of material			**0.035**	**0.256**	**0.256**	**0.144**	**0.006**
Cl=cw/n	Labour cost	Cw	$/h	25	25	25	25	25
	Production rate	n	units/h	2000	2000	2000	2000	2000
	C_l unit cost of labour			**0.0125**	**0.0125**	**0.0125**	**0.0125**	**0.0125**
Ct=ct*k/n	Tooling cost	Ct	$/set	50000	50000	50000	50000	50000
	Total production run	n	units	250000	250000	250000	250000	250000
	Tooling life	nt	units	75000	75000	75000	75000	75000
	Sets of tooling required	k		3.333	3.333	3.333	3.333	3.333
	C_t unit cost of tooling			**0.667**	**0.667**	**0.667**	**0.667**	**0.667**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	50000	50000	50000	50000	30000
	Capital write-off time	Five	years	23400	23400	23400	23400	23400
	Load fraction	L		1	1	1	1	1
	Load sharing fraction	q		1	1	1	1	1
	C_e unit cost of capital equipment			**0.0011**	**0.0011**	**0.0011**	**0.0011**	**0.0006**
Coh=coh/n	Factory overhead	coh	$/h	50	50	50	50	50
	Production rate	n	units/h	2000	2000	2000	2000	2000
	C_{oh} unit cost of factory overhead			**0.025**	**0.025**	**0.025**	**0.025**	**0.025**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**0.740**	**0.961**	**0.961**	**0.850**	**0.711**

Figure 23: Initial Economic Analysis Cont'd

INITIAL ECONOMIC ANALYSIS									
Formula	Cost Element	Variables	Units	LS04-02 Rubber Grip	LS04-03 Memory Foam	LS04-04 Nut and Bolt	LS05-01 USB Hub	LS05-02 USB Holder	LS05-03 USB Top Cap
	Material			Rubber	EVA	Steel	ABS	ABS	ABS
Cm=m*cm/1-f	Material cost	Cm	$/lb	0.5	10	1	1.75	1.75	1.75
	is scrap	f		0.05	0.05	0.05	0.05	0.05	0.05
	Mass of part	m	lb	0.04	0.05	0.06	0.00342	0.01	0.009
	C_m unit cost of material			**0.021**	**0.526**	**0.063**	**0.006**	**0.018**	**0.017**
Cl=cw/n	Labour cost	Cw	$/h	25	25	25	25	25	25
	Production rate	n	units/h	2000	2000	2000	2000	2000	2000
	C_l unit cost of labour			**0.0125**	**0.0125**	**0.0125**	**0.0125**	**0.0125**	**0.0125**
Ct=ct*k/n	Tooling cost	Ct	$/set	50000	50000	50000	50000	50000	50000
	Total production run	n	units	250000	250000	250000	250000	250000	250000
	Tooling life	nt	units	75000	75000	75000	75000	75000	75000
	Sets of tooling required	k		3.333	3.333	3.333	3.333	3.333	3.333
	C_t unit cost of tooling			**0.667**	**0.667**	**0.667**	**0.667**	**0.667**	**0.667**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	30000	30000	30000	40000	40000	40000
	Capital write-off time	Five	years	23400	23400	23400	23400	23400	23400
	Load fraction	L		1	1	1	1	1	1
	Load sharing fraction	q		1	1	1	1	1	1
	C_e unit cost of capital equipment			**0.0006**	**0.0006**	**0.0006**	**0.0009**	**0.0009**	**0.0009**
Coh=coh/n	Factory overhead	coh	$/h	50	50	50	50	50	50
	Production rate	n	units/h	2000	2000	2000	2000	2000	2000
	C_{oh} unit cost of factory overhead			**0.025**	**0.025**	**0.025**	**0.025**	**0.025**	**0.025**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**0.726**	**1.231**	**0.768**	**0.711**	**0.723**	**0.722**

Figure 24: Initial Economic Analysis Cont'd

BREAK-EVEN ANALYSIS

$$Q_b = \frac{f}{P - v}$$

Where, Q_b= Break-Even Point

f = Fixed Costs

v= Variable Costs

P= Sales Price

Table 17: OME and Breakeven Analysis

Parts	Part nos.	Material cost ($)	Labor cost ($)	Manufacturing cost ($)	Selling price	Break-even analysis (Units)
Fan	LS01-01	0.022	0.0125	0.066	1.728	29525
Fan Housing-Top	LS01-02	0.21	0.0125	0.63	2.015	27894
Fan Housing-Bottom	LS01-03	0.214	0.0125	0.642	1.85	30798
Customized Motor-Controller	LS01-04	0.368	0.0125	1.104	1.5	44663
Fan Housing Carrier	LS01-05	0.504	0.0125	1.512	1.95	34880
Magnetic Strip	LS01-06	0.013	0.0125	0.039	1.68	30221
Base Stand-Left	LS02-01	0.395	0.0125	1.185	1.59	42283
Base Stand-Right	LS02-02	0.395	0.0125	1.185	1.97	32000
Crossbar Support	LS02-03	0.144	0.0125	0.432	1.949	27894
Rise Stand-Left	LS02-04	0.035	0.0125	0.105	2.2	23229
Rise Stand-Right	LS02-05	0.035	0.0125	0.105	2	25608
Angle Stand-left	LS02-06	0.256	0.0125	0.768	1.861	31397
Angle Stand-Right	LS02-07	0.256	0.0125	0.768	1.861	31397
Phone Stand Holder	LS03	0.144	0.0125	0.432	1.85	29525
Hinge Rod	LS04-01	0.006	0.0125	0.018	1.711	29542
Rubber Grip	LS04-02	0.021	0.0125	0.063	1.726	29542
Memory Foam	LS04-03	0.526	0.0125	1.578	1.831	38685
Nut and Bolt	LS04-04	0.063	0.0125	0.189	1.8	28994
USB Hub	LS05-01	0.006	0.0125	0.018	1.711	29542
USB Holder	LS05-02	0.018	0.0125	0.054	1.75	29078
USB Top Cap	LS05-03	0.017	0.0125	0.051	1.722	29542
	Total	**3.648**	**0.2625**	**10.944**	**18**	

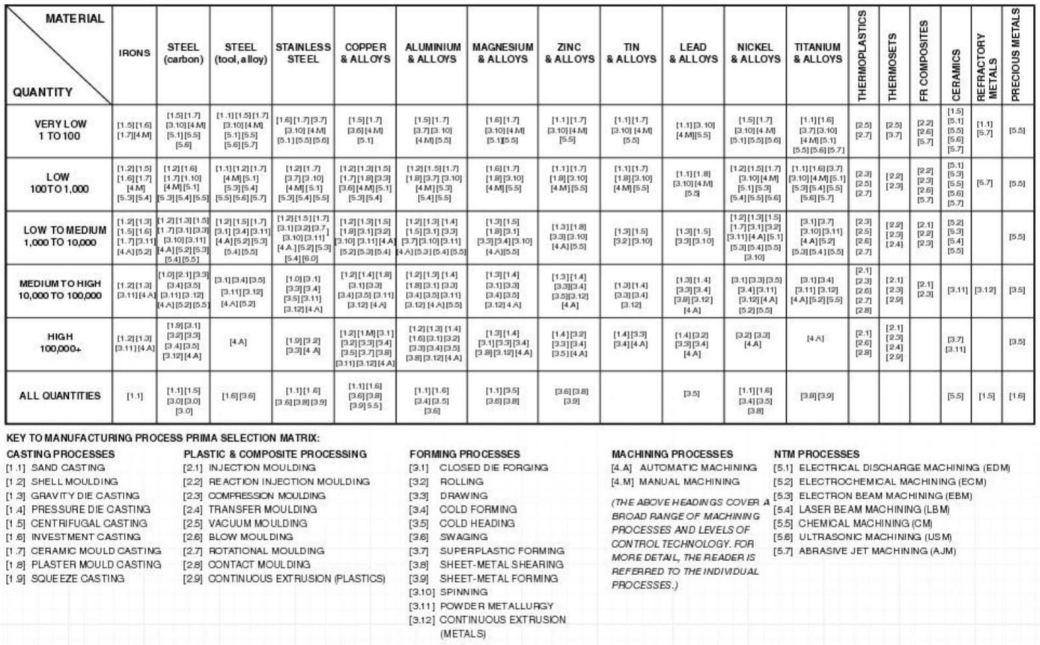
Table 18: Break-even Analysis for Product

Overall assembly unit	
Fixed cost ($)	**750000**
Variable cost ($)	**3.91**
Sales price Incl. Profit ($)	**40**
Break-even (Units)	**20781 units**

MANUFACTURING PROCESS ANALYSIS

The manufacturing process analysis for the laptop stand with a modular cooling pad highlights the strategic use of materials and manufacturing techniques to enhance performance and production efficiency. Injection molding is the primary method for producing complex plastic components such as fan housing and USB holders, chosen for their durability, ease of shaping, and suitability for high-volume production. Aluminum, used for structural components like the base stand and angle supports, is manufactured using die casting, which ensures high precision and strength while maintaining a lightweight design. Rubber, used for grips, is processed through compression molding to achieve flexibility and consistency. EVA foam, used for memory foam components, is shaped through thermoforming to ensure uniform thickness and ergonomic properties. Stainless steel hinge rods are precisely crafted through machining to provide durability and resistance to wear. These processes align with the functional demands of the product, emphasizing scalability, durability, and consistent quality.

Based on the material-process selection matrix, injection molding remains the optimal choice for ABS components like fan housing and USB holders due to its ability to produce intricate shapes efficiently in high volumes (economic batch size **10^3 to 10^7**). For aluminum parts such as the base and angle support, die casting is the preferred method for ensuring dimensional accuracy and consistency in medium-to-high batch production (**10^3 to 10^6**). Compression molding is well-suited for rubber grips due to its cost-effectiveness and simplicity, while thermoforming is ideal for EVA foam components, ensuring ergonomic performance. Machining is the optimal process for steel hinge rods, offering precision and strength, while wire drawing could be used to shape smaller steel parts if required consistently.



MATERIAL / QUANTITY	IRONS	STEEL (carbon)	STEEL (tool, alloy)	STAINLESS STEEL	COPPER & ALLOYS	ALUMINIUM & ALLOYS	MAGNESIUM & ALLOYS	ZINC & ALLOYS	TIN & ALLOYS	LEAD & ALLOYS	NICKEL & ALLOYS	TITANIUM & ALLOYS	THERMOPLASTICS	THERMOSETS	FR COMPOSITES	CERAMICS	REFRACTORY METALS	PRECIOUS METALS
VERY LOW 1 TO 100	[1.5] [1.6] [1.7] [4.M]	[1.5] [1.7] [3.10] [4.M] [5.1] [5.5] [5.6]	[1.1] [1.5] [1.7] [3.10] [4.M] [5.1] [5.5] [5.6] [5.7]	[1.6] [1.7] [3.7] [3.10] [4.M] [5.1] [5.5] [5.6]	[1.5] [1.7] [3.6] [4.M] [5.1]	[1.5] [1.7] [3.7] [3.10] [4.M] [5.5]	[1.6] [1.7] [3.10] [4.M] [5.1] [5.5]	[1.1] [1.7] [3.10] [4.M] [5.5]	[1.1] [1.7] [3.10] [4.M] [5.5]	[1.1] [3.10] [4.M] [5.5]	[1.5] [1.7] [3.10] [4.M] [5.1] [5.5] [5.6]	[1.1] [1.6] [3.7] [3.10] [4.M] [5.1] [5.5] [5.6] [5.7]	[2.5] [2.7]	[2.5] [3.7]	[2.2] [2.6] [5.7]	[1.5] [5.1] [5.5] [5.6] [5.7]	[1.1] [5.7]	[5.5]
LOW 100 TO 1,000	[1.2] [1.5] [1.6] [1.7] [4.M] [5.3] [5.4]	[1.2] [1.6] [1.7] [1.10] [4.M] [5.1] [5.3] [5.4] [5.5]	[1.1] [1.2] [1.7] [4.M] [5.1] [5.3] [5.4] [5.5] [5.6] [5.7]	[1.2] [1.7] [3.7] [3.10] [4.M] [5.1] [5.3] [5.4] [5.5]	[1.2] [1.3] [1.5] [1.7] [1.8] [3.7] [3.6] [4.M] [5.1] [5.3] [5.4]	[1.2] [1.5] [1.7] [1.8] [3.7] [3.10] [4.M] [5.3] [5.4] [5.5]	[1.6] [1.7] [1.8] [3.10] [4.M] [5.5]	[1.1] [1.7] [1.8] [3.10] [4.M] [5.5]	[1.1] [1.7] [1.8] [3.10] [4.M] [5.5]	[1.1] [1.8] [3.10] [4.M] [5.3]	[1.2] [1.5] [1.7] [3.10] [4.M] [5.1] [5.3] [5.4] [5.5] [5.6]	[1.1] [1.6] [3.7] [3.10] [4.M] [5.1] [5.3] [5.4] [5.5] [5.6] [5.7]	[2.3] [2.5] [2.7]	[2.2] [2.3]	[2.2] [2.3] [2.6] [5.7]	[5.1] [5.3] [5.5] [5.6] [5.7]	[5.7]	[5.5]
LOW TO MEDIUM 1,000 TO 10,000	[1.2] [1.3] [1.5] [1.6] [1.7] [3.11] [4.A] [5.2]	[1.2] [1.3] [1.5] [1.7] [3.1] [3.3] [3.10] [3.11] [4.A] [5.2] [5.3] [5.4] [5.5]	[1.2] [1.5] [1.7] [3.1] [3.4] [3.11] [4.A] [5.2] [5.3] [5.4] [5.5]	[1.2] [1.5] [1.7] [3.1] [3.3] [3.7] [3.10] [3.11] [4.A] [5.2] [5.3] [5.4] [6.0]	[1.2] [1.3] [1.5] [1.8] [2.1] [3.2] [3.10] [3.11] [4.A] [5.2] [5.3] [5.4]	[1.2] [1.3] [1.4] [1.5] [2.1] [3.2] [3.7] [3.10] [3.11] [4.A] [5.3] [5.4] [5.5]	[1.2] [1.3] [1.4] [1.8] [3.3] [3.4] [3.10] [4.A] [5.5]	[1.3] [1.8] [3.3] [3.10] [4.A] [5.5]	[1.3] [1.5] [3.2] [3.10]	[1.3] [1.5] [3.3] [3.10]	[1.2] [1.3] [1.5] [1.7] [3.1] [3.2] [3.11] [4.A] [5.1] [5.3] [5.4] [5.5] [5.6] [3.10]	[3.1] [3.7] [3.10] [3.11] [4.A] [5.2] [5.3] [5.4] [5.5]	[2.3] [2.5] [2.6] [2.7]	[2.2] [2.3] [2.4]	[2.1] [2.2] [2.3]	[5.2] [5.3] [5.4] [5.5]		[5.5]
MEDIUM TO HIGH 10,000 TO 100,000	[1.2] [1.3] [3.11] [4.A]	[1.0] [2.1] [3.3] [3.4] [3.5] [3.11] [3.12] [4.A] [5.2] [5.5]	[3.1] [3.4] [3.5] [3.11] [3.12] [4.A] [5.2]	[1.0] [3.1] [3.3] [3.4] [3.5] [3.11] [3.12] [4.A]	[1.2] [1.4] [1.8] [3.1] [3.3] [3.4] [3.5] [3.11] [3.12] [4.A]	[1.2] [1.3] [1.4] [1.8] [3.1] [3.3] [3.4] [3.5] [3.11] [3.12] [4.A] [5.5]	[1.3] [1.4] [3.1] [3.3] [3.4] [3.5] [3.12] [4.A]	[1.3] [1.4] [3.3] [3.4] [3.5] [3.8] [3.12] [4.A]	[1.3] [1.4] [3.2] [3.3]	[1.3] [1.4] [3.3] [3.4] [3.12]	[3.1] [3.3] [3.5] [3.4] [3.11] [3.12] [4.A] [5.2] [5.5]	[3.1] [3.4] [3.11] [3.12] [4.A] [5.2] [5.5]	[2.1] [2.3] [2.6] [2.7] [2.8]	[2.1] [2.3] [2.6]	[2.1] [2.3]	[3.11]	[3.12]	[3.5]
HIGH 100,000+	[1.2] [1.3] [3.11] [4.A]	[1.9] [3.1] [3.2] [3.3] [3.4] [3.5] [3.12] [4.A]	[4.A]	[1.9] [3.2] [3.3] [4.A]	[1.2] [1.M] [3.1] [3.2] [3.3] [3.4] [3.5] [3.7] [3.8] [3.11] [3.12] [4.A]	[1.2] [1.3] [1.4] [1.6] [3.1] [3.2] [3.3] [3.4] [3.5] [3.8] [3.12] [4.A]	[1.3] [1.4] [3.1] [3.3] [3.4] [3.8] [3.12] [4.A]	[1.4] [3.2] [3.3] [3.4] [3.5] [4.A]	[1.4] [3.3] [3.4] [4.A]	[1.4] [3.2] [3.3] [3.4] [4.A]	[3.2] [3.3] [4.A]	[4.A]	[2.1] [2.6] [2.8]	[2.1] [2.3] [2.4] [2.9]		[3.7] [3.11]		[3.5]
ALL QUANTITIES	[1.1]	[1.1] [1.5] [3.0] [3.1] [3.0]	[1.6] [3.6]	[1.1] [1.6] [3.6] [3.8] [3.9]	[1.1] [1.6] [3.6] [3.8] [3.9] [5.5]	[1.1] [1.6] [3.4] [3.5] [3.6]	[1.1] [3.5] [3.6] [3.8]	[3.6] [3.8] [3.9]		[3.5]	[1.1] [1.6] [3.4] [3.5] [3.8]	[3.8] [3.9]				[5.5]	[1.5]	[1.6]

KEY TO MANUFACTURING PROCESS PRIMA SELECTION MATRIX:

CASTING PROCESSES
[1.1] SAND CASTING
[1.2] SHELL MOULDING
[1.3] GRAVITY DIE CASTING
[1.4] PRESSURE DIE CASTING
[1.5] CENTRIFUGAL CASTING
[1.6] INVESTMENT CASTING
[1.7] CERAMIC MOULD CASTING
[1.8] PLASTER MOULD CASTING
[1.9] SQUEEZE CASTING

PLASTIC & COMPOSITE PROCESSING
[2.1] INJECTION MOULDING
[2.2] REACTION INJECTION MOULDING
[2.3] COMPRESSION MOULDING
[2.4] TRANSFER MOULDING
[2.5] VACUUM MOULDING
[2.6] BLOW MOULDING
[2.7] ROTATIONAL MOULDING
[2.8] CONTACT MOULDING
[2.9] CONTINUOUS EXTRUSION (PLASTICS)

FORMING PROCESSES
[3.1] CLOSED DIE FORGING
[3.2] ROLLING
[3.3] DRAWING
[3.4] COLD FORMING
[3.5] COLD HEADING
[3.6] SWAGING
[3.7] SUPERPLASTIC FORMING
[3.8] SHEET-METAL SHEARING
[3.9] SHEET-METAL FORMING
[3.10] SPINNING
[3.11] POWDER METALLURGY
[3.12] CONTINUOUS EXTRUSION (METALS)

MACHINING PROCESSES
[4.A] AUTOMATIC MACHINING
[4.M] MANUAL MACHINING
(THE ABOVE HEADINGS COVER A BROAD RANGE OF MACHINING PROCESSES AND LEVELS OF CONTROL TECHNOLOGY. FOR MORE DETAIL, THE READER IS REFERRED TO THE INDIVIDUAL PROCESSES.)

NTM PROCESSES
[5.1] ELECTRICAL DISCHARGE MACHINING (EDM)
[5.2] ELECTROCHEMICAL MACHINING (ECM)
[5.3] ELECTRON BEAM MACHINING (EBM)
[5.4] LASER BEAM MACHINING (LBM)
[5.5] CHEMICAL MACHINING (CM)
[5.6] ULTRASONIC MACHINING (USM)
[5.7] ABRASIVE JET MACHINING (AJM)

Figure 25: Influence of Materials on Process Selection

Figure 25 is a matrix used for selecting manufacturing processes based on the type of material and the quantity required. It categorizes materials such as iron, steel, aluminum, and various alloys, as well as plastics and composites, ceramics, and precious metals. For each material type, it suggests the most suitable manufacturing processes based on production volume (ranging from very low to high quantities). The chart also provides a selection of processes under various categories like casting, plastic processing, forming, machining, and non-traditional manufacturing methods (NTM). This matrix is designed to assist in determining the appropriate production methods tailored to specific materials and production scales. Below, Figure 26 gives an idea of the batch size of components for a particular manufacturing method to be commercially viable.

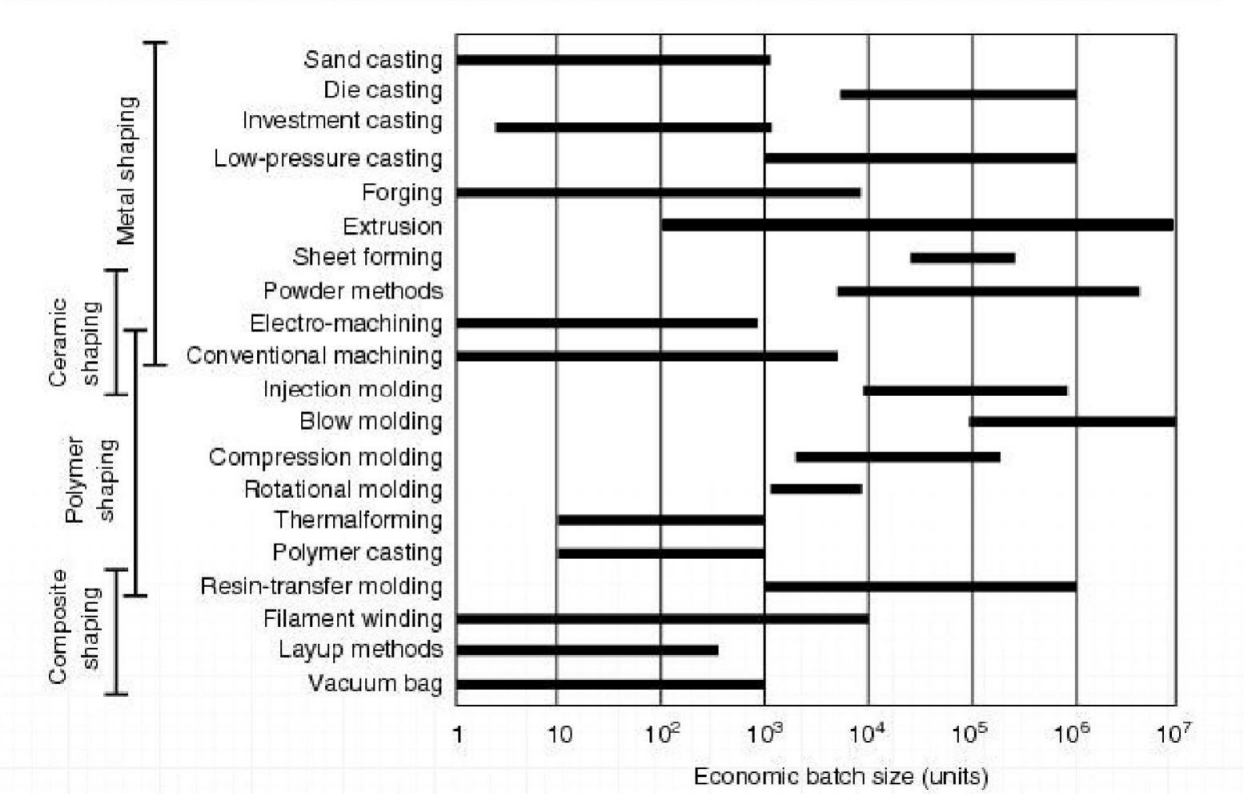


Figure 26: Manufacturing Process V/s Economic Batch Size

1. ABS Components (Fan Housing, USB Holder, USB Cap)

Material Overview: ABS is a thermoplastic known for its toughness, durability, and suitability for creating complex geometries. It is ideal for components like fan housing and USB holders.

Manufacturing Process:

- **Injection Molding:** Injection molding is the preferred process for ABS due to its ability to produce precise geometries, consistent surface finishes, and scalability for high production volumes.

Alternative:

- **Rotational Molding:** Suitable for more significant ABS parts with simpler geometries.

2. Aluminum Alloy Components (Base Stand, Rise Stand, Angle Supports)

Material Overview: Aluminum alloys are lightweight, corrosion-resistant, and strong, making them suitable for structural parts such as the base and supports.

Manufacturing Process:

- **Die Casting:** Die casting provides excellent dimensional accuracy, smooth surface finishes, and is effective for high-volume production.

Alternative:

- **Extrusion:** Useful for linear components like the crossbar supports, with secondary machining for precision.

3. Rubber Components (Rubber Grip)

Material Overview: Rubber is flexible, durable, and ideal for grips and ergonomic features.

Manufacturing Process:

- **Compression Molding:** Compression molding is suitable for producing rubber parts with simple shapes and consistent dimensions.

4. EVA Foam Components (Memory Foam)

Material Overview: EVA foam is lightweight, shock-absorbing, and resilient, ideal for padding or ergonomic applications.

Manufacturing Process:

- **Thermoforming:** Thermoforming ensures uniform thickness and properties, making it well-suited for producing foam components at medium-to-high volumes.

5. Stainless Steel Components (Hinge Rod)

Material Overview: Stainless steel is durable, corrosion-resistant, and strong, making it appropriate for high-stress parts like hinge rods.

Manufacturing Process:

- **Machining:** Machining ensures precision and consistency for small, high-strength components.

Summary Table of Processes

Component	Material	Primary Process	Reason
Fan Housing (Top/Bottom)	ABS	Injection Molding	High precision and scalability for complex geometries
Base Stand	Aluminum Alloy	Die Casting	Ensures dimensional accuracy and consistent quality
Rubber Grip	Rubber	Compression Molding	Effective for simple, flexible components
Hinge Rod	Stainless Steel	Machining	Delivers precision and durability for small components
Memory Foam	EVA Foam	Thermoforming	Produces uniform foam thickness for ergonomic applications

Conclusion

The manufacturing processes chosen for the laptop stand to ensure durability, precision, and scalability. Injection molding and die casting are ideal for high-volume production of ABS and aluminum components. In contrast, compression molding and machining are suited for rubber and steel parts requiring precision and flexibility. Thermoforming efficiently produces foam components for ergonomic applications, aligning with functional and production requirements.

DESIGN CHANGES

a. Integration of two fan housing

Before

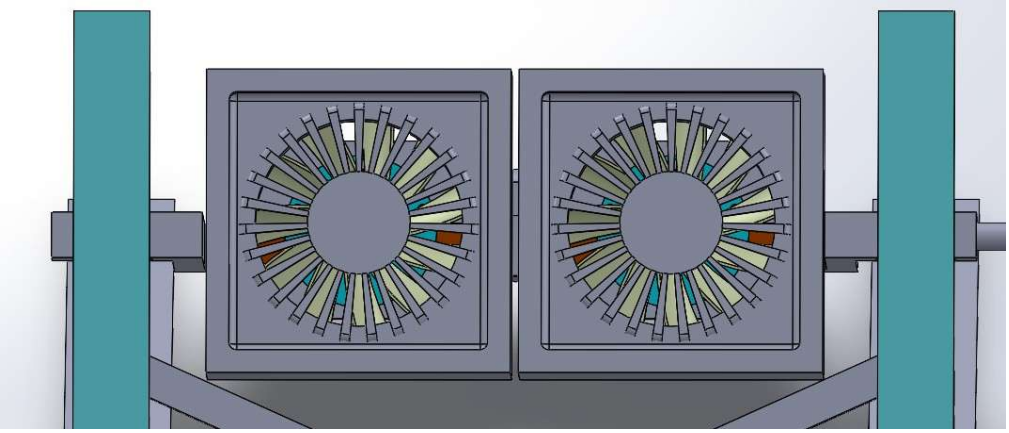


Figure 27: Modular Cooling Pad Before

After

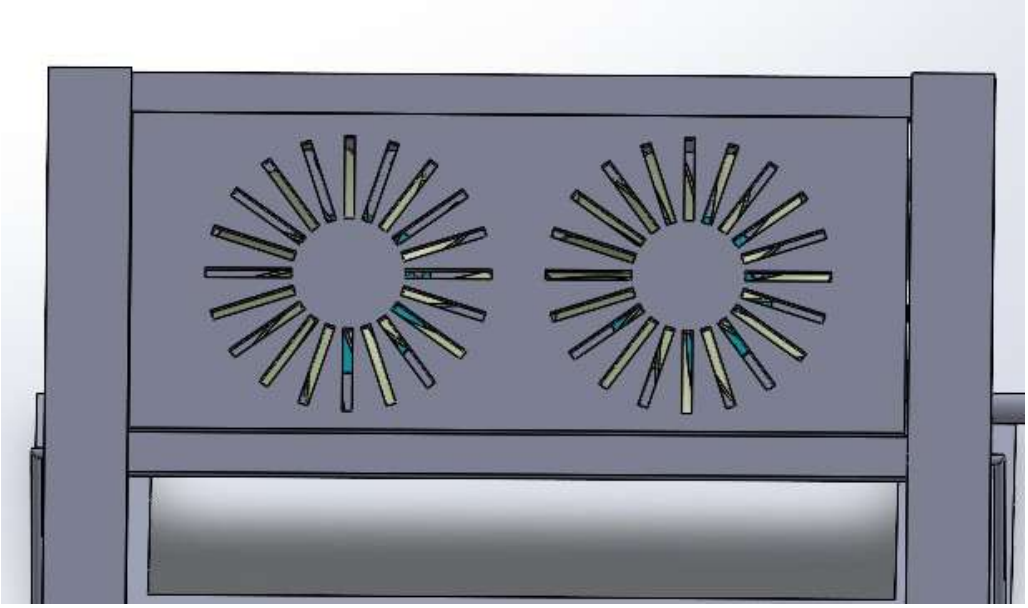


Figure 28: Modular Cooling Pad After

Explanation: While analyzing the initial design, we thought we could integrate both housings in a single box-type housing. The motor controller and fans are attached to the lower base part, i.e., the lower fan housing, through a bead lock. Also, there is enough space to place wires connecting the USB. No routing features are provided, as only a few wirings are required to power the fans. The setup is attached to the angle stand, **reducing three parts**.

b. Mobile stand

Before

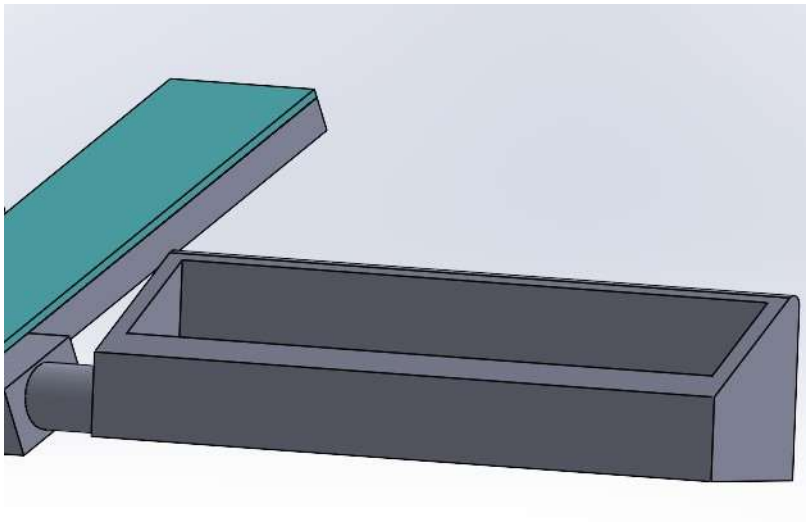


Figure 29: Mobile Stand before

After

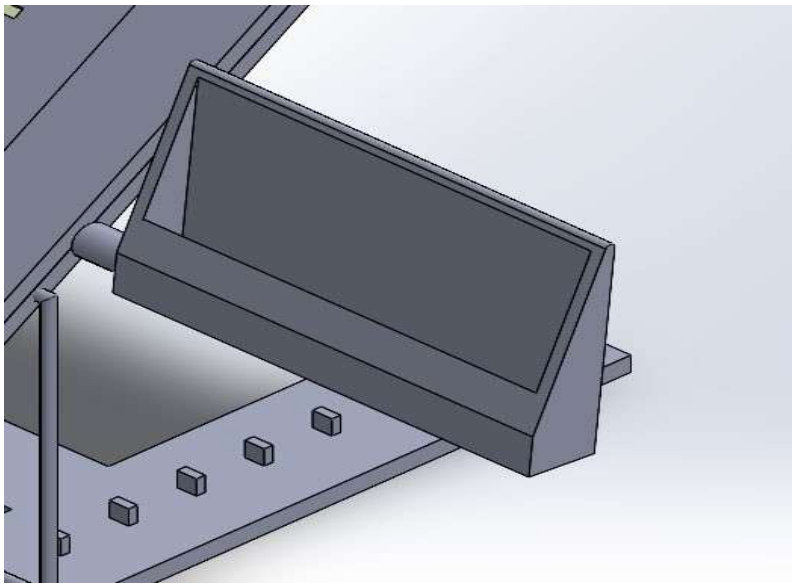


Figure 30: Mobile Stand After

Explanation: We benchmarked a few laptop stands that had mobile stands. The laptops tended to fall off the holder. We intend to increase the contact area to foolproof the holding.

c. Elimination of crossbar

Before

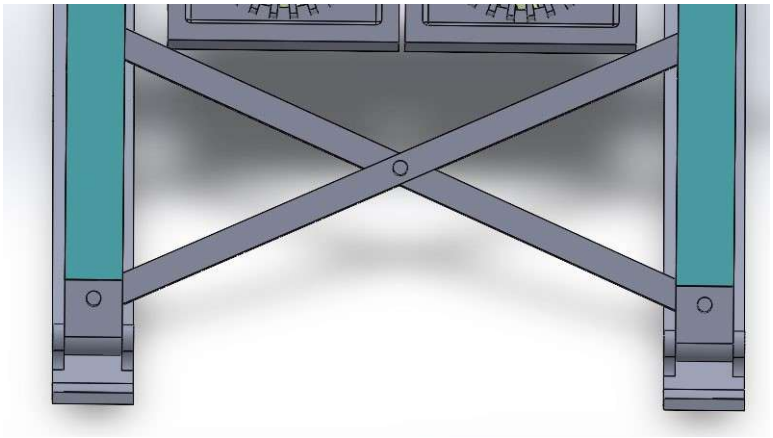


Figure 31: Crossbar Support Before

After

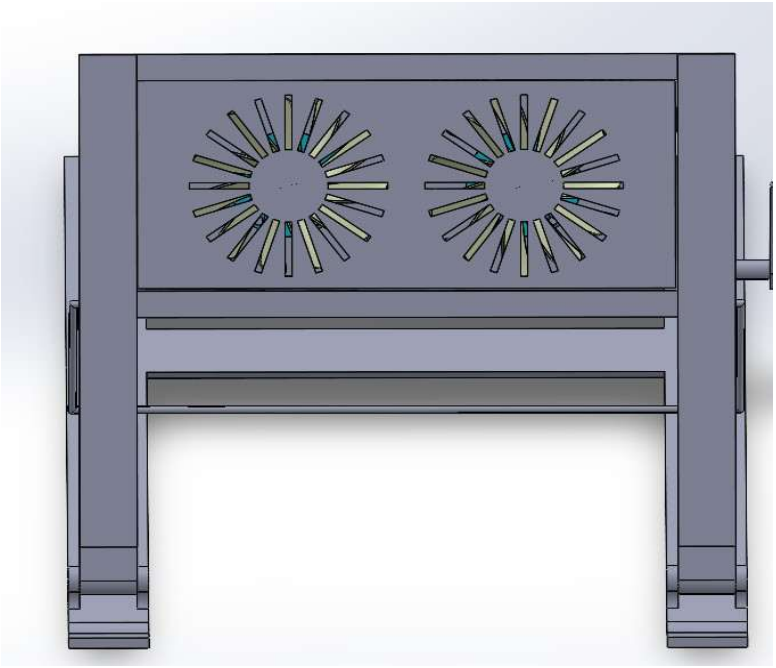


Figure 32:Crossbar Support After

Explanation: Cross bars are meant to expand and compress. In our concept, specifically targeted at laptops, we propose fixing the width and adding the modular cooling fan box at the modeled optimal position to get the best cooling effect. So, we eliminated two bars across each other to keep the width fixed.

d. Modification of Rise stand

Before

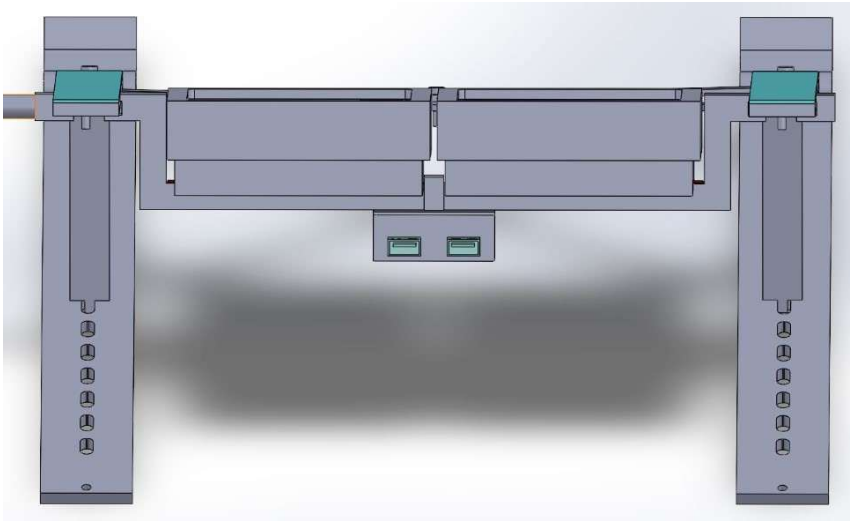


Figure 33: Rise Stand Before

After

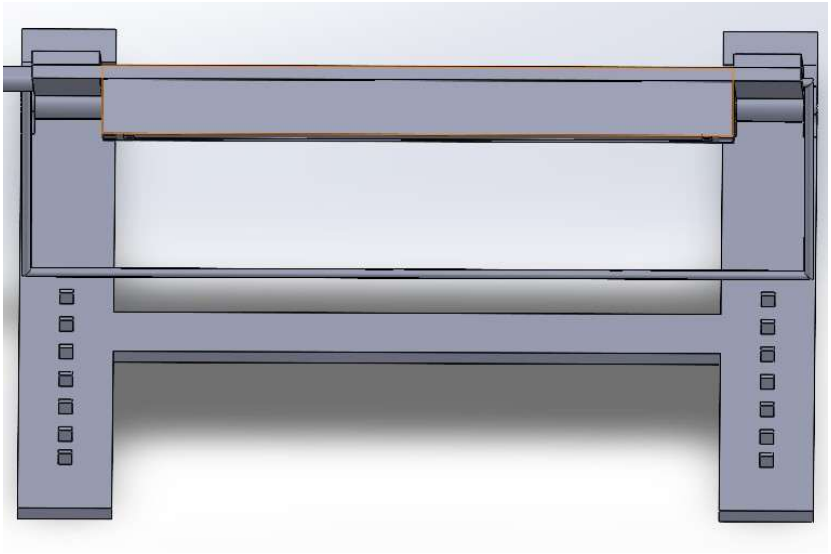


Figure 34: Rise Stand After

Explanation: Earlier, we had slots in the base stand to accommodate the rise stand projections. Intending to reduce the material, we propose using a tubular rise stand that is easy to attach to counterparts and rotate. Parallelly, the few projections are raised in the base stand, as shown, to hold the angle stands at the position.

e. Elimination of USB assembly

Before

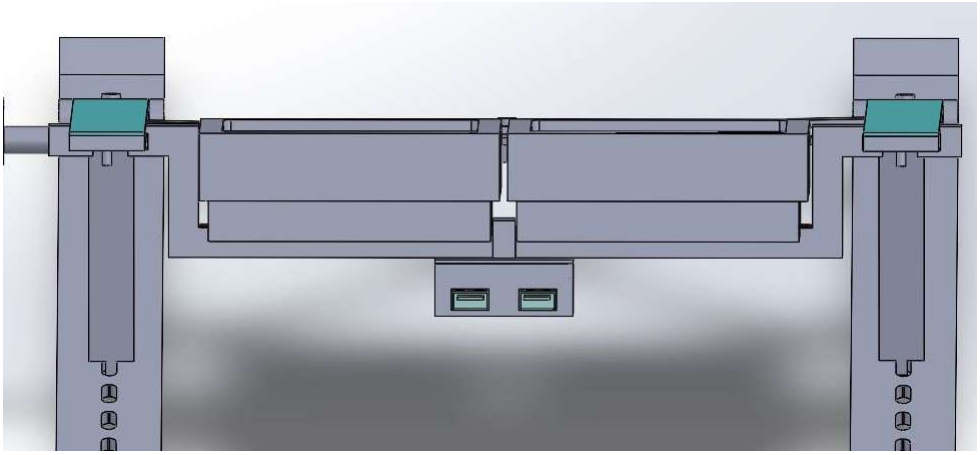


Figure 35: USB Module Hub Before

After

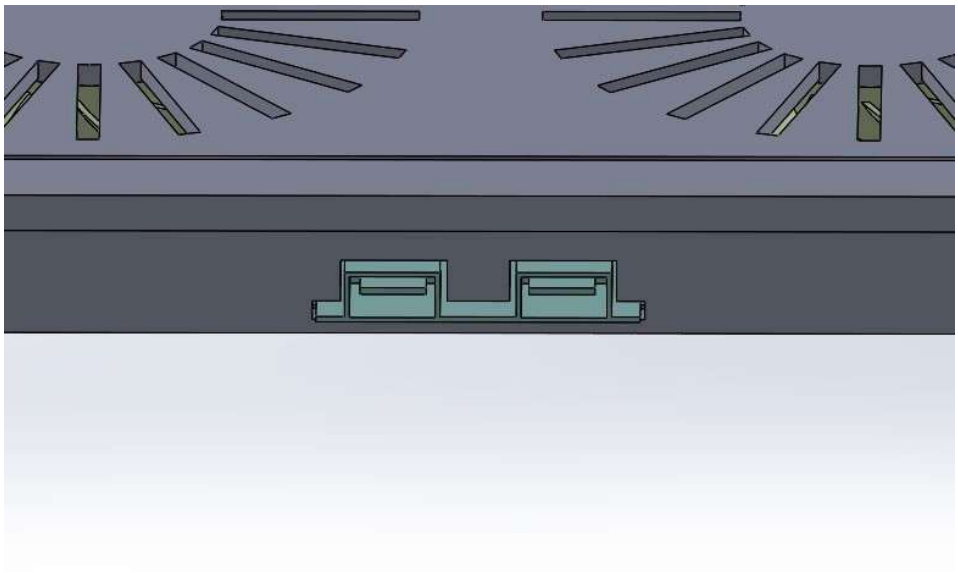


Figure 36: USB Module After

Explanation: As a benefit of integrating the housing to hold two fans, there was enough space to hold the USB hub. We propose to place the hub attached to the lower fan housing through a positive lock that is projected outside to access it. This design requires a slot in the upper fan housing to allow the USB outside. It's an undercut for that injection molded part. However, with the help of a slider, we try to overcome the issue. This mold complexity is a trade-off we would like to take as it's cost-effective, too (avoids another part/mold).

NEW DFA ANALYSIS

DFA Analysis Worksheet																									
Assembly Name: Laptop stand with modular cooling pad									Team: Group 6								Date: 11/19/2024								
If the answer is Yes to any of the metrics or questions enter a 1. If the answer is No then enter 0. Each cell must have a number.																									
Part		Complexity		Functional Analysis / Redesign Opportunity				Error Proofing		Handling			Insertion				Secondary Operations								
Part Number	Part Name	Number of Parts (N_p)	Number of Interfaces (N_i)	Theoretical Minimum Part	Part Can Be Standardized (if not already standard)	Cost (Low/Medium/High)	Practical Minimum Part	Assemble Wrong Part/ Omit Part	Assemble Part Wrong Way Around	Tangle, Nest, or Stick Together	Flexible, Fragile, Sharp or Slippery	Pliers, Tweezers, or Magnifying Glass Needed	Difficult to Align/ Locate	Holding Down Required	Resistance to Insertion	Obstructed Access/ Visibility	Re-orient Workpiece	Screw, Drill, Twist, Rivet, Bend, or Crimp	Weld, Solder, or Glue	Paint, Lube, Heat, Apply Liquid or Gas	Test, Measure or Adjust				
LS01	FAN ASSEMBLY																								
LS01-01	Fan	2	1	1	1	M	1	0	0	1	1	0	0	0	0	0	0	0	0	0	0				
LS01-02	Fan Housing-Top	2	4	1	1	M	1	0	0	1	0	0	0	0	1	0	0	0	0	0	0				
LS01-03	Fan Housing-Bottom	2	4	1	1	M	1	0	0	1	0	0	0	0	1	0	0	0	0	0	0				
LS01-04	Customised Motor-Controller	1	2	1	1	L	1	0	0	1	0	0	0	0	0	0	0	1	0	0	1				
LS02	BASE ASSEMBLY																								
LS02-01	Base Stand	1	2	1	1	L	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0				
LS02-02	Rise Stand	1	5	1	1	L	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0				
LS02-03	Angle Support	1	4	1	1	L	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0				
LS03	Mobile Stand	1	1	1	0	L	1	0	0	0	0	0	0	0	1	0	0	0	0	1	0				
LS04	OFF THE SHELF																								
LS04-01	Hinge rod	2	2	1	1	L	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0				
LS04-02	Rubber grip	2	1	1	1	L	1	0	0	1	1	0	0	0	0	0	0	0	1	1	0				
LS04-03	Memory foam	2	2	1	1	L	1	1	0	1	1	0	0	0	0	0	0	0	1	1	0				
LS04-04	Rubber grip Base	4	1	1	1	L	1	0	0	1	0	0	0	0	0	0	0	0	1	0	0				
LS05	USB Module	1	1	1	1	M	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
Totals		22	30	13	12	0	13	1	0	7	3	0	0	0	4	0	0	1	3	6	1				
Design for Assembly Metrics		25.69		59.1%	←Theor. Effy. Pract. Effy.→		59.1%	0.08		0.77			0.31				0.85								
Targets		30																							

Figure 37: Final DFA Analysis

We achieved our target and even exceeded it, reducing the DFA metrics score by approximately 23 points. The new DFA metric score was 25.69. We reduced the parts to 22 and the interfaces to 30. We achieved this by redesigning the stand into three parts and the cooling pad. Both theoretical and practical efficiencies improved to 59.1% from 45.7% and 40%, respectively.

NEW MATERIAL AND MANUFACTURING PROCESS ANALYSIS

After completing the final design model for the laptop stand with a modular cooling pad, the materials and manufacturing processes underwent a thorough review to ensure their suitability for performance, cost, and production efficiency. Following the detailed evaluations, the initial material selections and manufacturing methods were primarily retained, with minimal changes to optimize functionality and scalability. Below is a comprehensive summary of the final choices.

Materials

1. **ABS Plastic**
    - **Application:** Fan housing, USB holder, and USB top cap.

- **Reasons for Retention:** ABS plastic remained preferred due to its excellent impact resistance, dimensional stability, and ease of molding into complex shapes. Its ability to withstand mechanical stress while maintaining a lightweight structure made it ideal for structural and aesthetic components of the laptop stand.
- **Changes:** None.

2. **Aluminum Alloy**
   - **Application:** Base stand, rise stands, angle supports, and crossbar support.
   - **Reasons for Retention:** Aluminum was chosen for its high strength-to-weight ratio, corrosion resistance, and ease of processing through die casting. Its durability ensures long-term performance under load-bearing conditions.
   - **Changes:** No material changes, but the geometry of certain parts was slightly adjusted to reduce weight without compromising strength.
3. **Rubber**
   - **Application:** Rubber grip pads for the base.
   - **Reasons for Retention:** Rubber provides flexibility, anti-slip properties, and excellent shock absorption, ensuring stability during use. Its cost-effectiveness and manufacturability through compression molding made it ideal for this application.
   - **Changes:** None.
4. **EVA Foam**
   - **Application:** Memory foam used for ergonomic padding.
   - **Reasons for Retention:** EVA foam remained a practical choice for its lightweight, shock-absorbing properties and comfort. Its thermoformability allows for precise shaping and consistent thickness.
   - **Changes:** None.
5. **Stainless Steel**
   - **Application:** Hinge rods for structural connections.
   - **Reasons for Retention:** Stainless steel was retained for its high mechanical strength, corrosion resistance, and machinability, critical for precision components like hinge rods.
   - **Changes:** None.

Manufacturing Processes

1. **Injection Molding (ABS Plastic Components)**
    - **Process Summary:** Injection molding produced parts like the fan housing, USB holders, and USB top caps. This process ensured high precision, scalability, and consistent quality for large production runs.
    - **Changes:** Minor tool adjustments were made to accommodate slight geometric refinements.
2. **Die Casting (Aluminum Alloy Components)**
    - **Process Summary:** Aluminium parts, such as the base, was manufactured using die casting. This process provided excellent dimensional accuracy and surface finish, ensuring structural integrity.
    - **Changes:** None.
3. **Compression Molding (Rubber Grip Pads)**
    - **Process Summary:** Rubber grip pads were manufactured through compression molding, which enabled efficient production of flexible, durable parts.
    - **Changes:** None.
4. **Thermoforming (EVA Foam)**
    - **Process Summary:** Thermoforming was used to create the ergonomic foam padding. This method allowed for precise control over thickness and shape, ensuring uniformity and comfort.
    - **Changes:** None.
5. **Machining (Stainless Steel Hinge Rods)**
    - **Process Summary:** Hinge rods were produced through automated machining processes to achieve precise dimensions and high strength.
    - **Changes:** None.

Final Adjustments and Considerations

1. **Geometry Optimization:** The final model incorporated minor geometric adjustments to reduce weight, improve manufacturability, and streamline assembly. For example:
    - The base stand was slightly hollowed out in non-critical areas to minimize material usage without compromising strength.
    - Projections were added to facilitate easier attachment of the modular cooling pad.

2. **Sustainability:** No major changes were made to the material choices, but the design aimed to minimize waste during production. Processes like injection molding and die casting were optimized to reduce scrap material.
3. **Assembly and Integration:** The cooling pad's modular nature was retained, and the assembly process was not changed. Alignment features were included to simplify attachment and improve user experience.
4. **Cost Efficiency:** The project ensured consistency in cost estimates by retaining the original materials and processes. High-volume production techniques, like injection molding and die casting, maintained scalability and affordability.

Final Remarks:

The final model of the laptop stand with a modular cooling pad retained most of its original materials and manufacturing processes, demonstrating their effectiveness in meeting functional, aesthetic, and economic goals. Minimal adjustments to part geometry enhanced weight reduction and manufacturability, while the modular design ensured user convenience. By optimizing the production methods, the design achieved scalability and sustainability, making it ready for large-scale manufacturing with minimal waste.

NEW ECONOMIC ANALYSIS

A new economic analysis was done on the redesigned model to observe whether the part redesigns and reductions have reduced manufacturing costs. Later, an OME analysis and break-even analysis were done to observe how the number of units needed to become profitable changed.

Assumptions made in analysis:

For individual components, a fixed cost of $50000 was used. This cost includes manufacturing, depreciation, and other costs. We assumed a fixed cost of $750000 for the entire assembly as those costs typically include marketing, inventory, shipping, and logistics. For capital write-off time in the sheet, we assumed operations for 45 hours each for 52 weeks, and we amortized the components over 5 and 10 years based on their complexity and size. When calculating their break-even points, we assumed some profit on all the individual components. We also assumed a total product profit of $22 for product break-even analysis. With these assumptions, we did the economic analysis calculations visible below and the break-even analysis, the latter shown in Tables 19 and 20 below.

FORMULAS

ECONOMIC ANALYSIS

- $C_m=\frac{m*c_m}{1-f}$, where **C_m= Unit cost of material**, **m= mass of part, c_m=cost of material, f=fraction of material scrap**.

- $C_L=\frac{c_{ww}}{n'}$, where **C_L= Unit Labor Cost, c_w= hourly cost of wages and benefits, n' =** ppppppppppppppppppp pprrpprr.
- $C_T=\frac{ct\cdot k}{n}$ where, **C_T= Tooling Cost, c_t= cost of making tools, n= production run, k= tooling wear factor** (**k= n**/by life of tooling raised to next highest integer).
- $C_E=(\frac{1}{n'})\cdot(\frac{c_e}{Lt_{wwo}})\cdot q$, **where c_e= cost of capital equipment, n'= production rate, L= Load Factor, t_{wo}= capital write-off time, q= fraction of equipment sharing between products.**
- $C_{OH}=\frac{cOOH}{n'}$, **where c_{OH}= overhead hourly rate, n'= production rate**.
- **C_U= Cm+C_L+C_T+C_E+C_{OH}**

These calculations are shown in Figures 38,39 and 40 below.

FINAL ECONOMIC ANALYSIS

Formula	Cost Element	Variables	Units	LS01-01 Fan	LS01-02 Fan Housing-Top	LS01-03 Fan Housing-Bottom	LS01-04 Customised Motor-Controller
	Material			ABS	ABS	ABS	ABS
Cm=m*cm/1-f	Material cost	Cm	$/lb	1.75	1.75	1.75	1.75
	Fraction of process that is scrap	f		0.05	0.05	0.05	0.05
	Mass of part	m	lb	0.012	0.114	0.1164	0.2
	C_m unit cost of material			**0.022**	**0.210**	**0.214**	**0.368**
Cl=cw/n	Labour cost	Cw	$/h	25	25	25	25
	Production rate	n	units/h	2000	2000	2000	2000
	C_l unit cost of labour			**0.0125**	**0.0125**	**0.0125**	**0.0125**
Ct=ct*k/n	Tooling cost	Ct	$/set	50000	50000	50000	50000
	Total production run	n	units	250000	250000	250000	250000
	Tooling life	nt	units	75000	75000	75000	75000
	Sets of tooling required	k		3.333	3.333	3.333	3.333
	C_t unit cost of tooling			**0.667**	**0.667**	**0.667**	**0.667**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	60000	60000	60000	60000
	Capital write-off time	Five	years	23400	23400	23400	23400
	Load fraction	L		1	1	1	1
	Load sharing fraction	q		1	1	1	1
	C_e unit cost of capital equipment			**0.00128**	**0.00128**	**0.00128**	**0.00128**
Coh=coh/n	Factory overhead	coh	$/h	50	50	50	50
	Production rate	n	units/h	2000	2000	2000	2000
	C_{oh} unit cost of factory overhead			**0.025**	**0.025**	**0.025**	**0.025**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**0.728**	**0.915**	**0.920**	**1.074**

Figure 38: Final Economic Analysis

FINAL ECONOMIC ANALYSIS

Formula	Cost Element	Variables	Units	LS02-01 Base Stand	LS02-02 Rise Stand	LS02-03 Angle Stand	LS03 Phone Stand Holder
	Material			Aluminium	Aluminium	Aluminium	ABS
Cm=m*cm/1-f	Material cost	Cm	$/lb	1.16	1.16	1.16	1.75
	Fraction of process that is scrap	f		0.05	0.05	0.05	0.05
	Mass of part	m	lb	0.3234	0.02842	0.20972	0.0784
	C_m unit cost of material			**0.395**	**0.035**	**0.2561**	**0.1444**
Cl=cw/n	Labour cost	Cw	$/h	25	25	25	25
	Production rate	n	units/h	2000	2000	2000	2000
	C_l unit cost of labour			**0.0125**	**0.0125**	**0.0125**	**0.0125**
Ct=ct*k/n	Tooling cost	Ct	$/set	50000	50000	50000	50000
	Total production run	n	units	250000	250000	250000	250000
	Tooling life	nt	units	75000	75000	75000	75000
	Sets of tooling required	k		3.333	3.333	3.333	3.333
	C_t unit cost of tooling			**0.667**	**0.667**	**0.667**	**0.667**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	50000	50000	50000	50000
	Capital write-off time	Five	years	23400	23400	23400	23400
	Load fraction	L		1	1	1	1
	Load sharing fraction	q		1	1	1	1
	C_e unit cost of capital equipment			**0.00107**	**0.00107**	**0.00107**	**0.00107**
Coh=coh/n	Factory overhead	coh	$/h	50	50	50	50
	Production rate	n	units/h	2000	2000	2000	2000
	C_{oh} unit cost of factory overhead			**0.025**	**0.025**	**0.025**	**0.025**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**1.100**	**0.740**	**0.961**	**0.850**

Figure 39: Final Economic Analysis-Cont'd

FINAL ECONOMIC ANALYSIS

Formula	Cost Element	Variables	Units	LS04-01 Hinge Rod	LS04-02 Rubber Grip	LS04-03 Memory Foam	LS05-01 USB Module
	Material			Stainless Steel	Rubber	EVA	ABS
Cm=m*cm/1-f	Material cost	Cm	$/lb	1	0.5	10	1.75
	Fraction of process that is scrap	f		0.05	0.05	0.05	0.05
	Mass of part	m	lb	0.0058	0.04	0.05	0.0028
	C_m unit cost of material			**0.0061**	**0.0211**	**0.5263**	**0.0052**
Cl=cw/n	Labour cost	Cw	$/h	25	25	25	25
	Production rate	n	units/h	2000	2000	2000	2000
	C_l unit cost of labour			**0.0125**	**0.0125**	**0.0125**	**0.0125**
Ct=ct*k/n	Tooling cost	Ct	$/set	50000	50000	50000	50000
	Total production run	n	units	250000	250000	250000	250000
	Tooling life	nt	units	75000	75000	75000	75000
	Sets of tooling required	k		3.333	3.333	3.333	3.333
	C_t unit cost of tooling			**0.667**	**0.667**	**0.667**	**0.667**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	30000	30000	30000	40000
	Capital write-off time	Five	years	23400	23400	23400	23400
	Load fraction	L		1	1	1	1
	Load sharing fraction	q		1	1	1	1
	C_e unit cost of capital equipment			**0.00064**	**0.00064**	**0.00064**	**0.00085**
Coh=coh/n	Factory overhead	coh	$/h	50	50	50	50
	Production rate	n	units/h	2000	2000	2000	2000
	C_{oh} unit cost of factory overhead			**0.025**	**0.025**	**0.025**	**0.025**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**0.711**	**0.726**	**1.231**	**0.710**

Figure 40: Final Economic Analysis-Cont'd

BREAK-EVEN ANALYSIS

$$Q_b = \frac{f}{P - v}$$

Where, Q_b= Break-Even Point

f = Fixed Costs

v= Variable Costs

P= Sales Price

Table 19: Final Design OME and Break-Even Analysis

Parts	Part nos.	Material cost ($)	Labor cost ($)	Manufacturing cost ($)	Selling price ($)	Break-even analysis (Units)
Fan	LS01-01	0.022	0.0125	0.066	1.728	29525
Fan Housing-Top	LS01-02	0.622	0.0125	1.866	2.5	21004
Fan Housing-Bottom	LS01-03	0.155	0.0125	0.465	.85	29718
Customized Motor-Controller	LS01-04	0.368	0.0125	1.104	1.5	44663
Base Stand	LS02-01	1.031	0.0125	3.093	4	16912
Rise Stand	LS02-02	0.746	0.0125	2.238	3	22306
Angle Stand	LS02-03	0.031	0.0125	0.093	.861	27510
Phone Stand Holder	LS03	0.1227	0.0125	0.3681	.945	24816
Hinge Rod	LS04-01	0.0061	0.0125	0.0183	.711	29544
Rubber Grip	LS04-02	0.0211	0.0125	0.0633	.320	46887
Memory Foam	LS04-03	0.5263	0.0125	1.5789	1.7	43059
USB Hub	LS05	0.0052	0.0125	0.0156	1	29528
	Total	**3.6564**	**0.15**	**10.9692**	**12.25**	

Table 20: New Breakeven Analysis for Product

Overall assembly unit	
Fixed cost ($)	**750000**
Variable cost ($)	**3.81**
Sales price Incl. Profit ($)	**35**
Break-even (Units)	**24043**

ETHICAL CONSIDERATION AND SAFETY

Ethical considerations and safety play a critical role in ensuring the product's usability, sustainability, and reliability when designing a laptop stand with a modular cooling pad. The materials selected should be non-toxic, recyclable, and environmentally sustainable to minimize ecological impact and ensure the safety of users. For example, avoiding harmful substances such as BPA in plastics or volatile compounds in adhesives promotes user health and safety. From a structural perspective, the design must prioritize stability and load-bearing capacity to prevent laptop tipping or accidental damage. Sharp edges or components that may cause injuries during use or assembly should be eliminated. The cooling pad must also comply with electrical safety standards to prevent overheating, short circuits, or fire hazards during operation. Adhering to ergonomic principles is equally important, ensuring that the stand alleviates strain on the user’s posture and promotes long-term comfort. Finally, ethical responsibility requires fair labor practices and sustainable sourcing of materials during production, ensuring that the product is manufactured under conditions that respect human rights and minimize environmental harm.

CONCLUSIONS

The development project of the **Laptop Stand with Modular Cooling Pad** was an in-depth process that enabled a comprehensive understanding of the product's design, material selection, and manufacturing processes. The primary objective was to create a functional, ergonomic, and cost-effective product while optimizing assembly efficiency and exploring sustainable design options. Throughout the project, we utilized benchmarking studies, iterative CAD modeling, and Design for Assembly (DFA) analyses to refine the design and ensure its feasibility for large-scale production.

A key focus of the project was to balance performance and manufacturability while minimizing costs. Material selection was critical as we evaluated alternatives for the base stand, cooling pad housing, and rubber grips. For example, ABS plastic, aluminum alloys, and EVA foam were confirmed as the optimal materials due to their strength, durability, and ease of processing. High-volume manufacturing techniques such as injection molding, die casting, and compression molding were retained for their scalability and precision, with minor adjustments to tooling and geometry to enhance efficiency.

The DFA analysis identified opportunities to simplify the assembly process and reduce part counts. Notable improvements included consolidating frame components into fewer parts, relocating the USB hub for better accessibility, and redesigning the attachment mechanism for the modular cooling pad. These changes reduced assembly complexity, minimized potential errors, and improved overall manufacturability without compromising structural integrity or performance.

This project also emphasized ethical considerations, including ensuring user safety, promoting sustainability through material choices, and adhering to ergonomic principles for enhanced user comfort. The modular design allowed users flexibility, enabling easy maintenance and customization. Furthermore, the design aligns with modern environmental standards by integrating sustainable materials and reducing production waste.

Ultimately, this project offered valuable insights into an innovative product's design and manufacturing processes. It served as an excellent opportunity for the team to apply engineering principles, learn collaborative problem-solving, and address real-world design challenges. The experience has not only resulted in a market-ready product but has also provided the team with a deeper understanding of how to innovate effectively in product development.

ACKNOWLEDGEMENTS

The team would like to thank Dr. Dan Riffell for his guidance throughout the project. He taught us the concepts behind DFMA and gave us all insight into the difference between good and bad engineering.

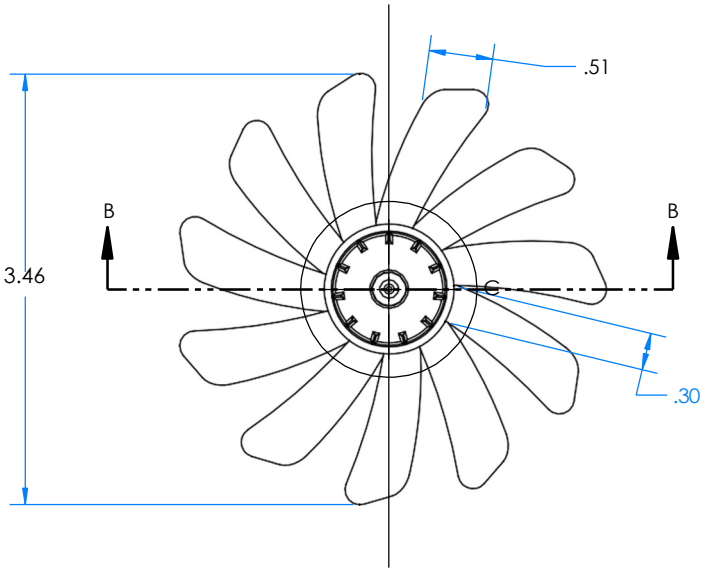
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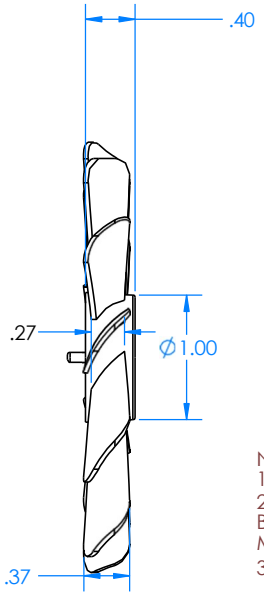
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[3].Squall 300 RGB Cooling Pad by Archer Tech Labs
[4].MOFT Laptop Stand for Desk
[5].Adjustable Laptop Riser Stand

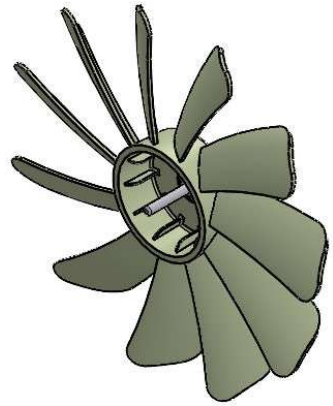
APPENDICES

- Design Change Drawings with Assembly and BOM.

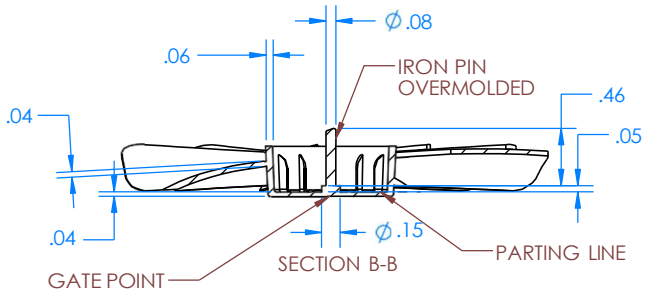
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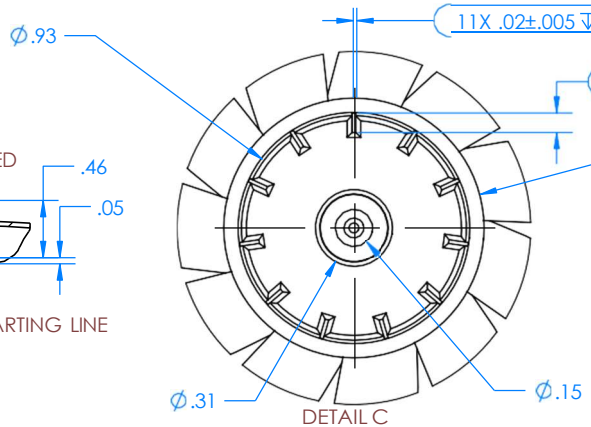






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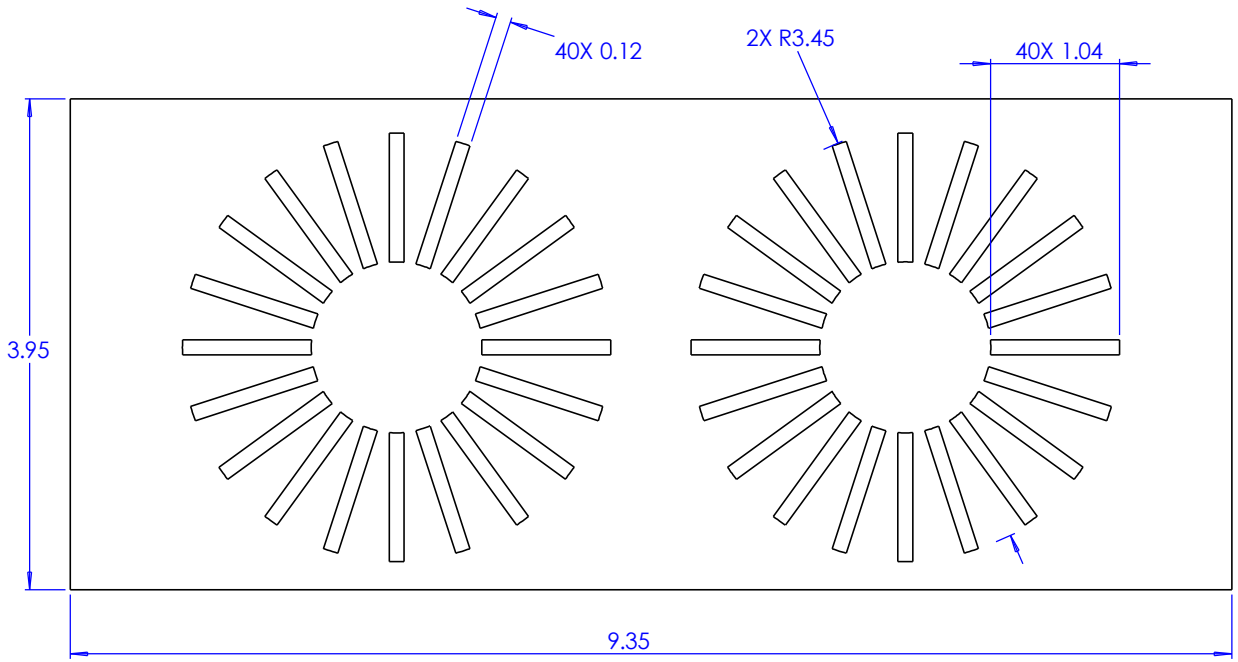


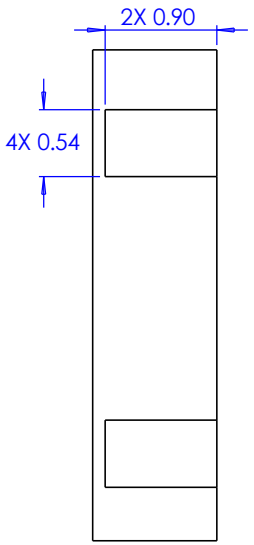
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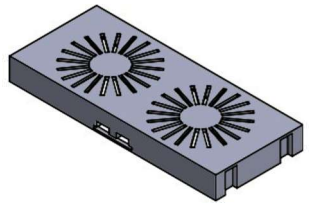
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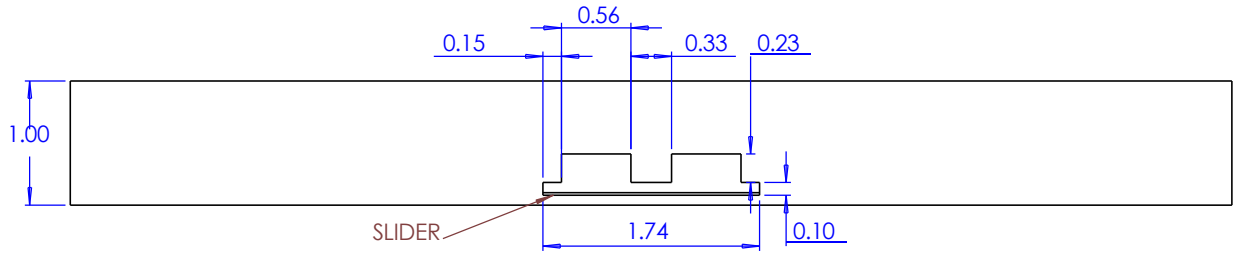
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SCALE 1:4

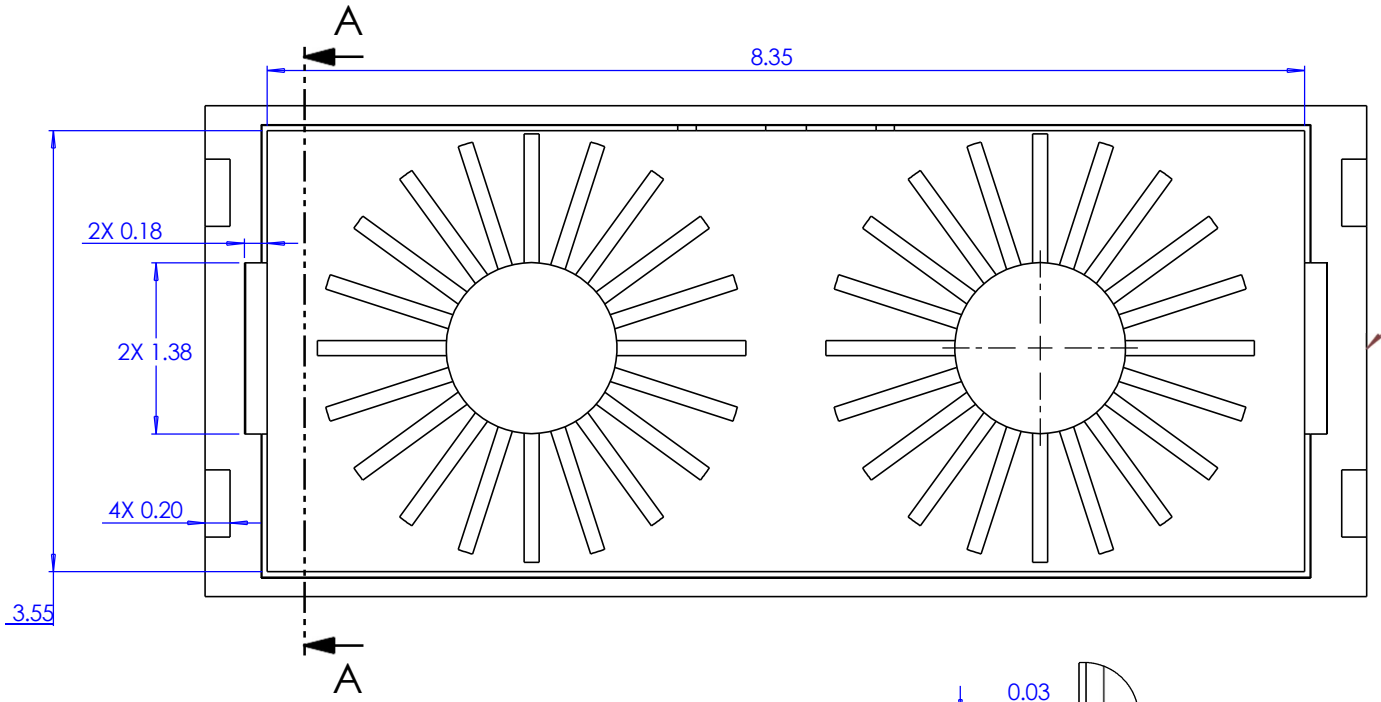


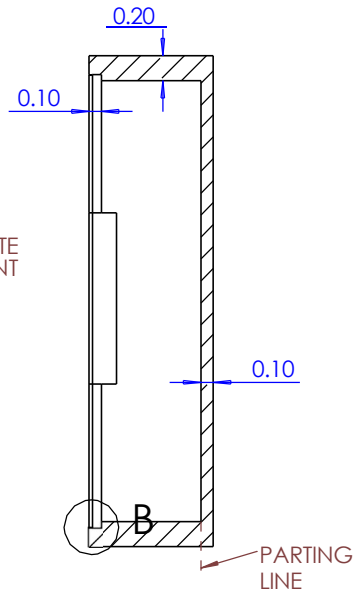
NOTES:
1. PART SHALL BE VISIBLY CLEAN OF ALL DIRT, OIL, AND SOLVENTS.
2. THIS IS A LIMITED DIMENSION DRAWING. UNSPECIFIED DIMENSIONS SHALL BE TAKEN FROM THE ATTACHED COMPUTER-GENERATED SOLID MODEL LS01-02 FAN HOUSING TOP REV B
3. CRITICAL OR INSPECTION DIMENSIONS ARE ENCLOSED WITH THE SYMBOL

TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ± 0.050 X.XX ± 0.010 X.XXX ± 0.005 X.X ° ± 0.5°	UNIVERSITY OF COLORADO 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427
MATERIAL: ABS		DESCRIPTION: FAN HOUSING TOP
FINISH: N/A	PN: LS01-02 REV: B	SHEET: 1 of 2

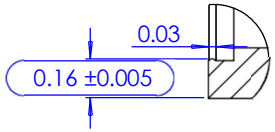
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SECTION A-A
SCALE 1 : 1



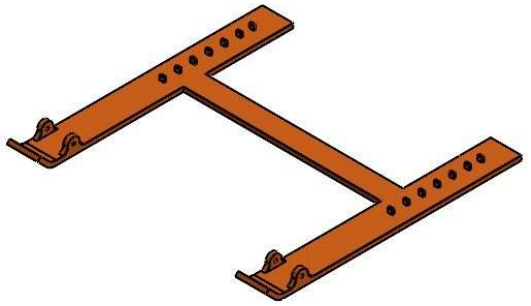
DETAIL B
SCALE 2 : 1

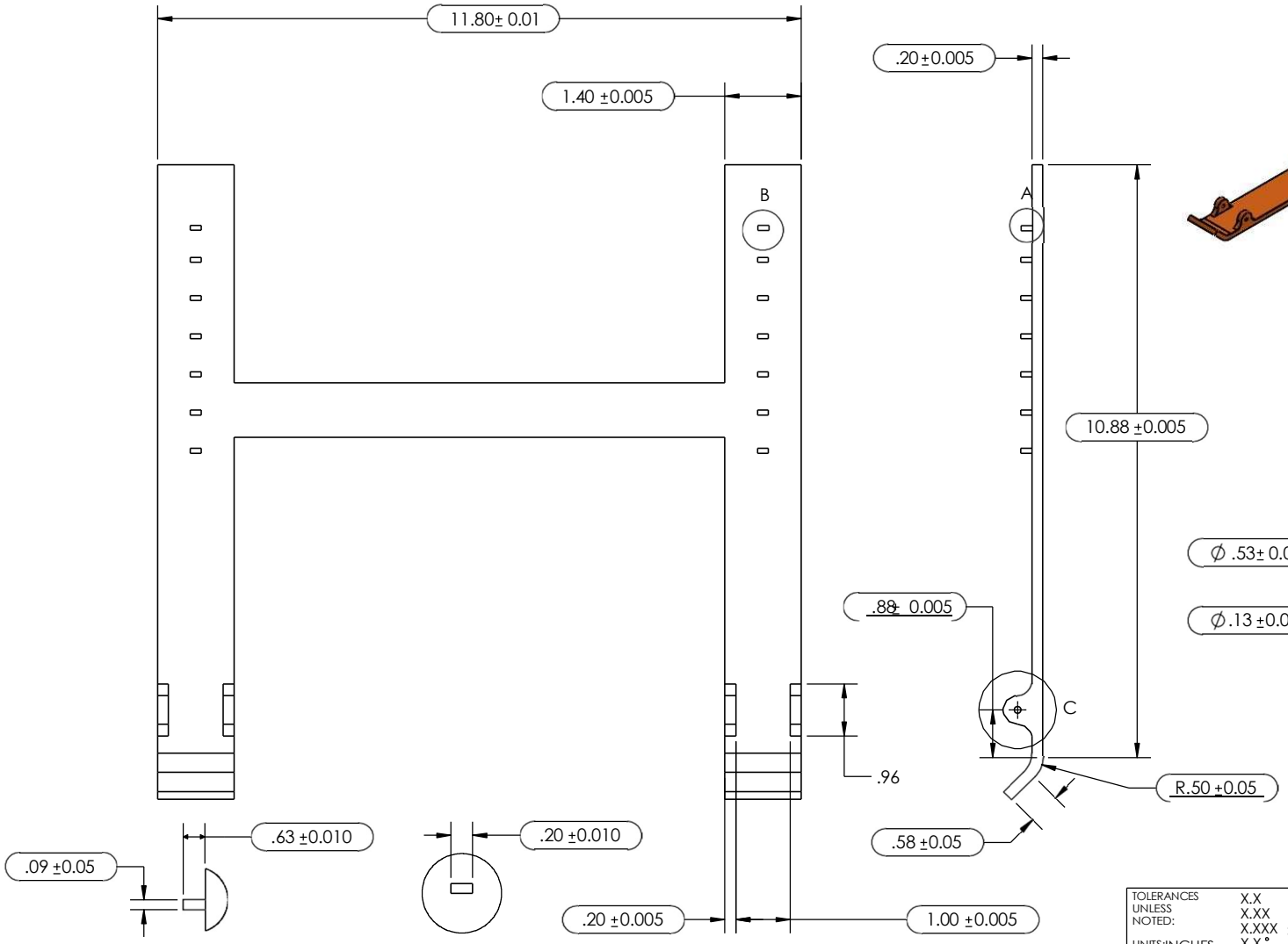
TOLERANCES UNLESS NOTED:	X.X ± 0.050 X.XX ± 0.010 X.XXX ± 0.005	UNIVERSITY OF COLORADO 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
UNITS:	X.X ° ± 0.5°		
MATERIAL ABS		DESCRIPTION FAN HOUSING TOP	
FINISH N/A		PN LS01-02	REV B
			SHEET 2 of 2

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REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	11/24/2024

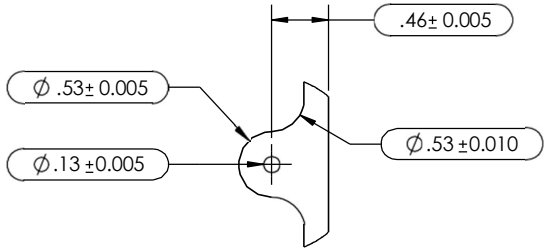




DETAIL A
SCALE 1 : 1

DETAIL B
SCALE 1 : 1

NOTE:
1.UNLESS MENTIONED SCALE 1:2



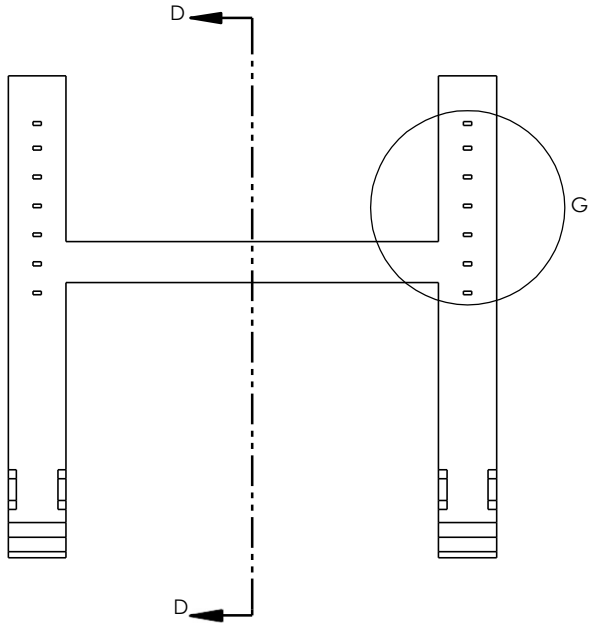
DETAIL C
SCALE 1 : 1

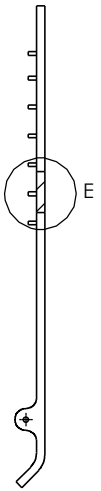
TOLERANCES UNLESS NOTED: X.X ±0.05, X.XX ±0.010, X.XXX ±0.005, X.X ° °1"	UNIVERSITY OF COLORADO BOULDER		
UNITS:INCHES			
MATERIAL: ALUMINIUM	DESCRIPTION: BASE STAND		
FINISH: N/A	PN: LS02-01	REV: A	SHEET: 1 of 2

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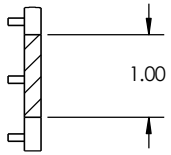
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REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	11/24/2024

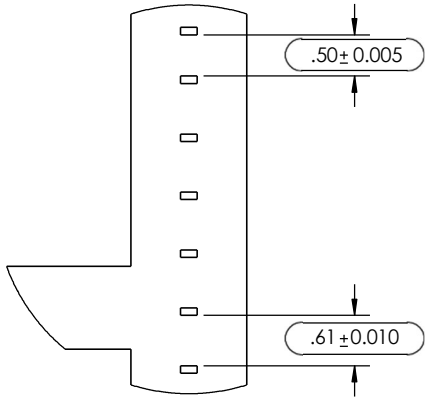




SECTION D-D
SCALE 1 : 3



DETAIL E
SCALE 2 : 3



DETAIL G
SCALE 2 : 3

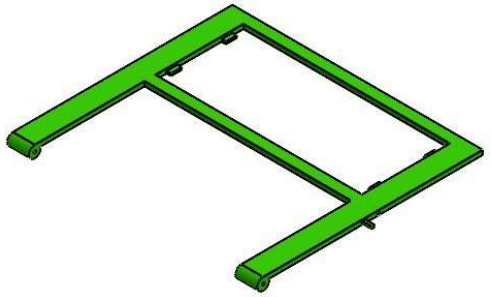
NOTE:
1. PART IS SYMMETRIC
2. THERE ARE 6 FEATURES IN 'G' FROM BOTTOM WHICH ARE SPACED .61" APART

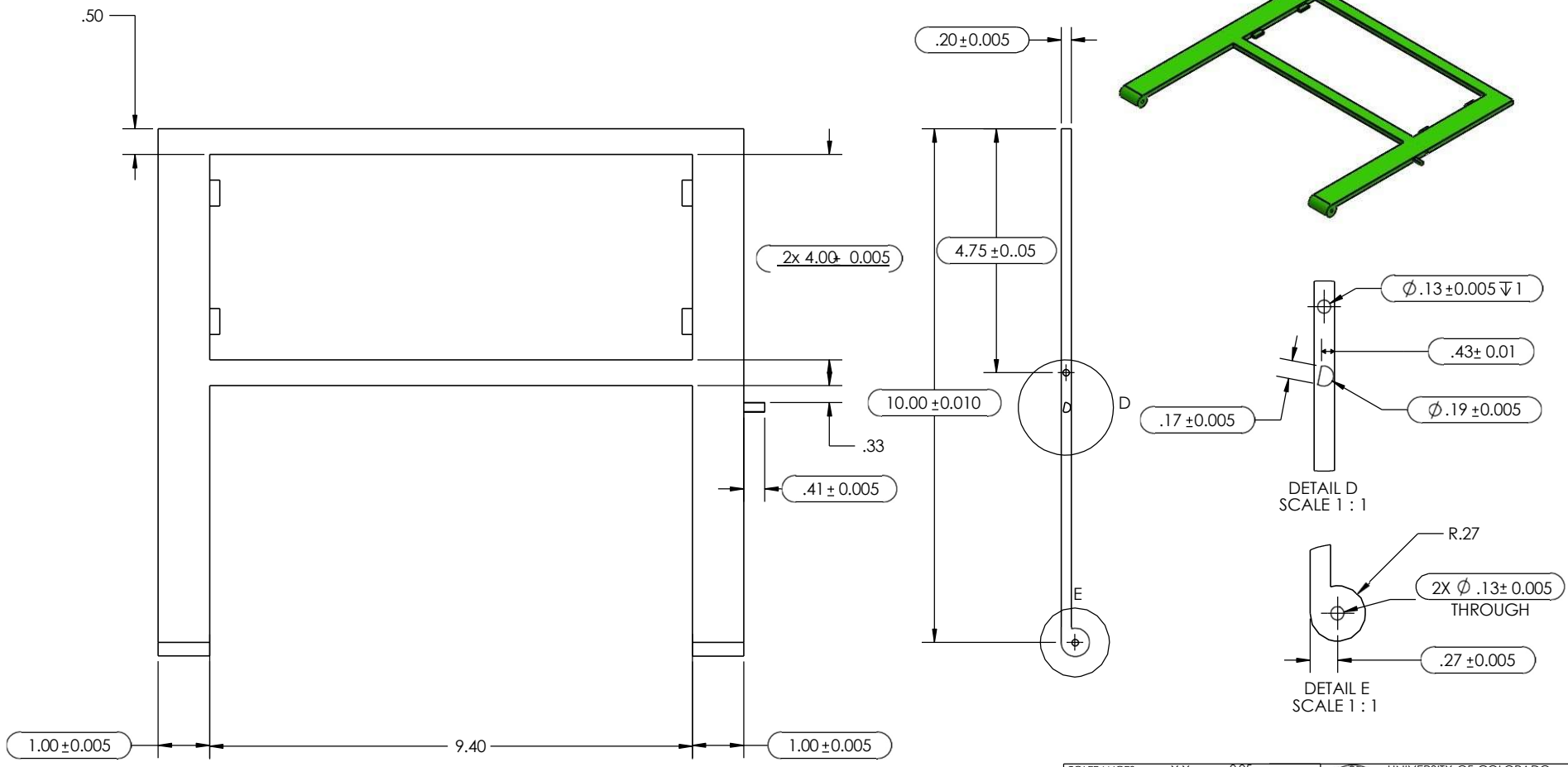
TOLERANCES UNLESS NOTED: UNITS:INCHES	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005 X.X ° °1"	UNIVERSITY OF COLORADO BOULDER	
MATERIAL: ALUMINIUM		DESCRIPTION: BASE STAND	
FINISH: N/A		PN: LS02-01	REV: A
			SHEET: 2 of 2

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REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	11/23/2024





DETAIL D
SCALE 1 : 1

DETAIL E
SCALE 1 : 1

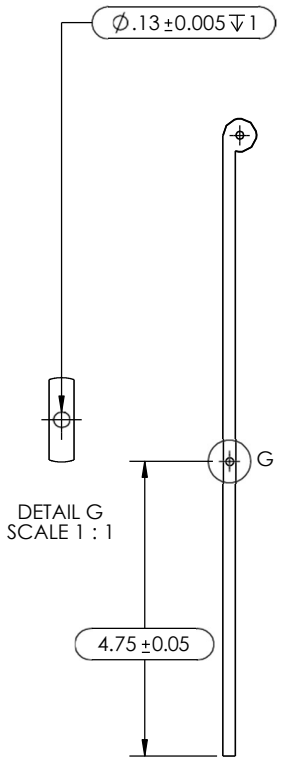
NOTE:
1.UNLESS MENTIONED SCALE 1:2

TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005 X.X ° °1"	UNIVERSITY OF COLORADO BOULDER
MATERIAL: ALUMINIUM		DESCRIPTION: RISE STAND
FINISH: N/A	PN: LS02-02	REV: A
		SHEET: 1 of 2

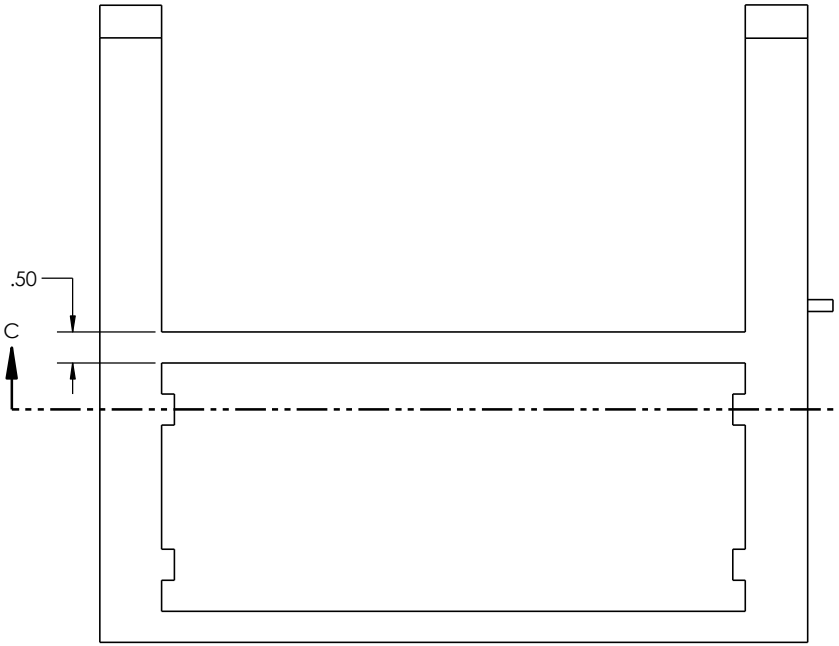
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.

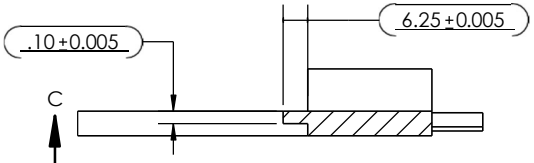
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REV	DESCRIPTION	DATE
A	INITIAL RELEASE	11/23/2024



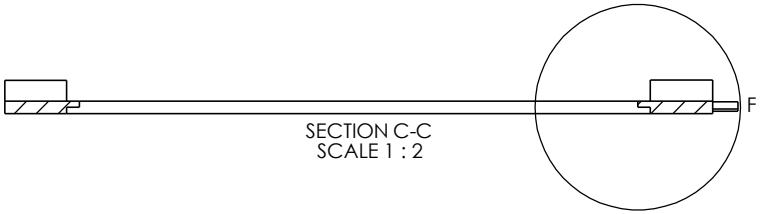
DETAIL G
SCALE 1 : 1





DETAIL F
SCALE 1 : 1

NOTE:
1.UNLESS MENTIONED SCALE 1:2



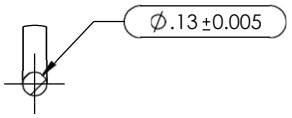
SECTION C-C
SCALE 1 : 2

TOLERANCES UNLESS NOTED:	X.X ± 0.05 X.XX ±0.010 X.XXX ±0.005	UNIVERSITY OF COLORADO BOULDER
UNITS:INCHES	X.X ° °1"	
MATERIAL: ALUMINIUM		DESCRIPTION: RISE STAND
FINISH: N/A	PN: LS02-04 / REV: A	SHEET: 2 of 2

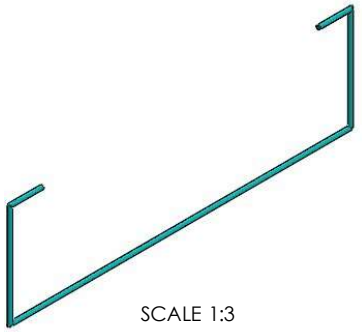
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UNIVERSITY OF COLORADO
BOULDER **SOLIDWORKS Educational Product. For Instructional Use Only.**

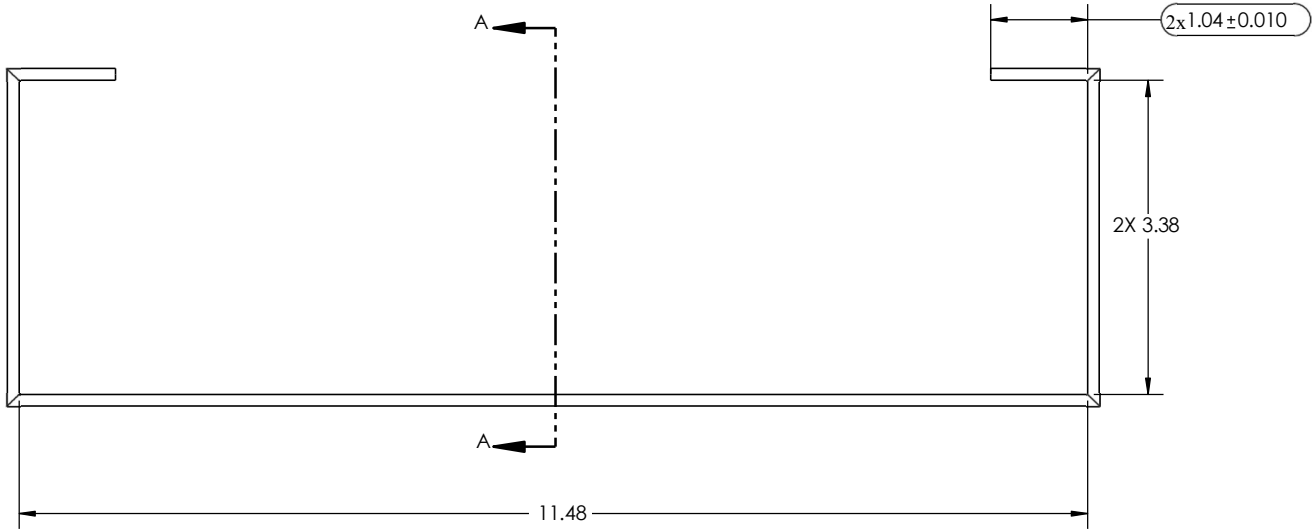
REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	11/24/2024

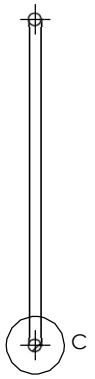


DETAIL C
SCALE 3 : 2



SCALE 1:3





SECTION A-A
SCALE 1.5 : 2

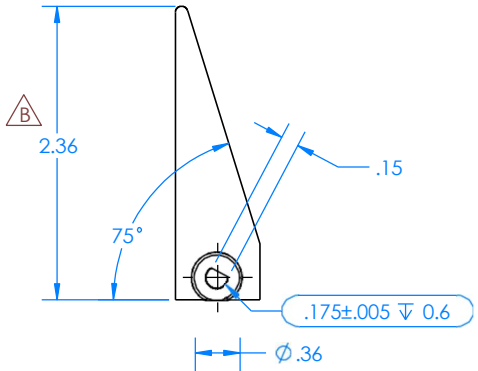
NOTE:
1. UNLESS MENTIONED SCALE 1.5:2
2. UNIFORM WALL DIAMETER OF .13" THROUGHOUT THE PART

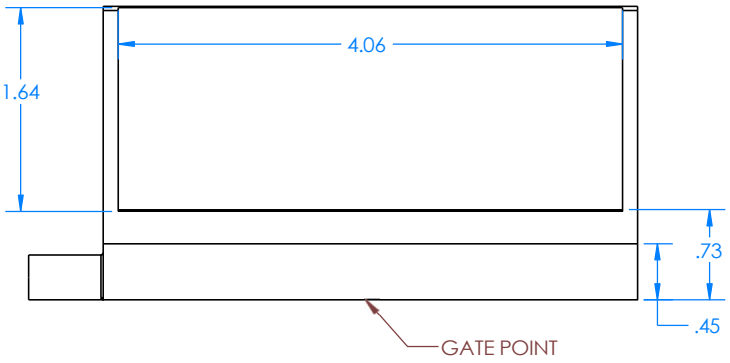
TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ±0.05 X.XX ± 0.010 X.XXX ± 0.005 X.X ° °1"	UNIVERSITY OF COLORADO BOULDER
MATERIAL: ALUMINIUM		DESCRIPTION: ANGLE SUPPORT
FINISH: N/A	PN: LS02-03	REV: A / SHEET: 1 of 1

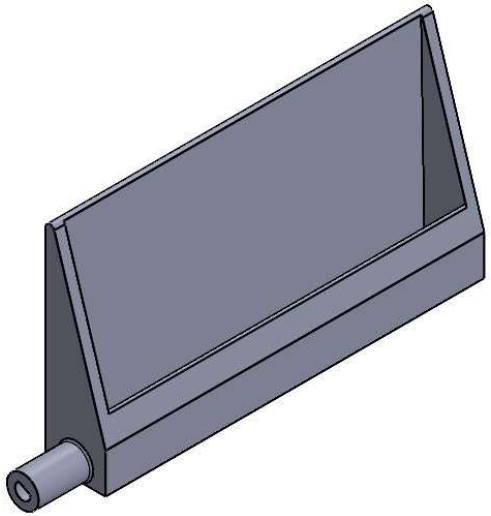
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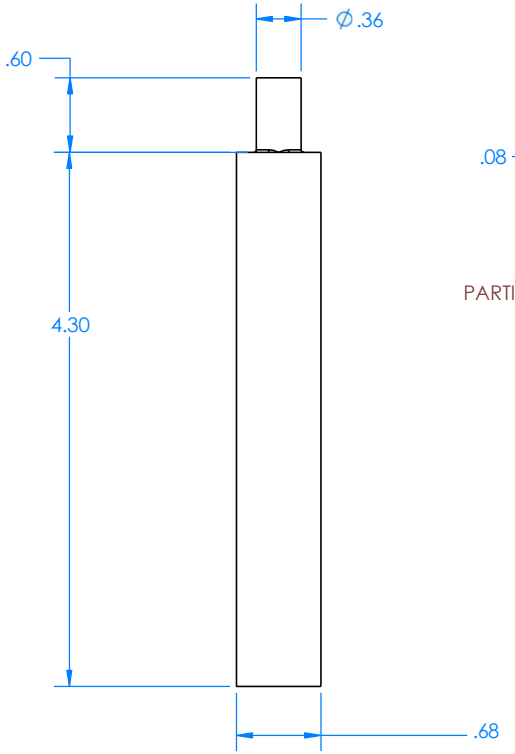
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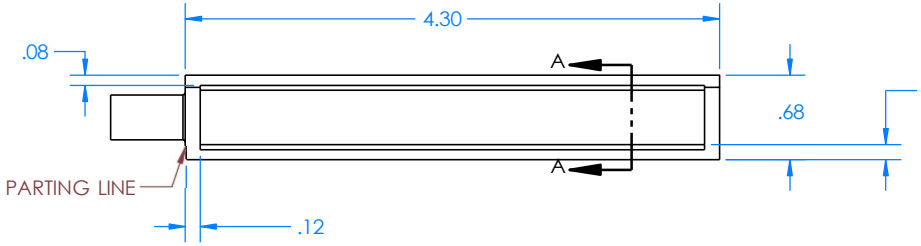
REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	DRAWN BY
	A	INITIAL RELEASE	11/22/2024	SARAVANA
6A	B	MOBILE SUPPORT SURFACE HEIGHT INCREASED	29/11/2024	SARAVANA

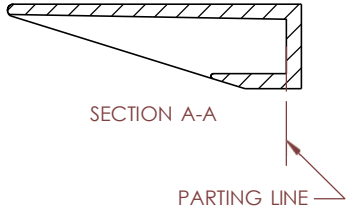












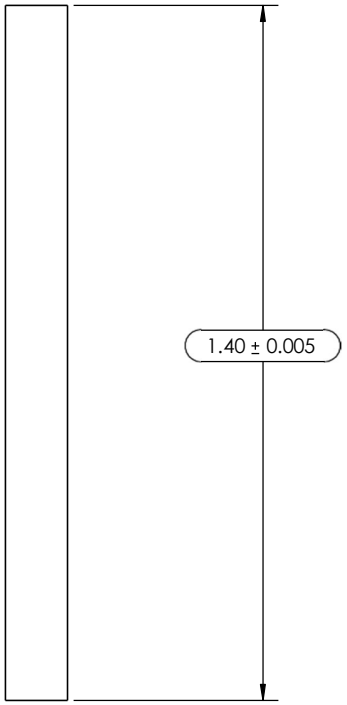
NOTES:
1. PART SHALL BE VISIBLY CLEAN OF ALL DIRT, OIL, AND SOLVENTS.
2. THIS IS A LIMITED DIMENSION DRAWING. UNSPECIFIED DIMENSIONS SHALL BE TAKEN FROM THE ATTACHED COMPUTER-GENERATED SOLID MODEL LS03 MOBILE STAND REV B
3. CRITICAL OR INSPECTION DIMENSIONS ARE ENCLOSED WITH THE SYMBOL

TOLERANCES UNLESS NOTED: UNITS:INCHES	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005 X.X ° 1°	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
MATERIAL: ABS		DESCRIPTION: MOBILE STAND	
FINISH: N/A	SCALE: 1:1	PN: LS03	REV: B
			SHEET: 1 of 1

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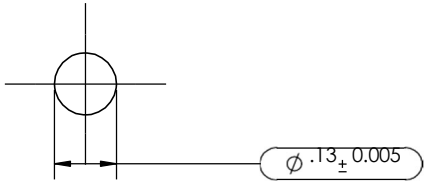
REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	11/24/2024





SCALE 2:1

NOTE:
1.UNLESS METIONED SCALE IS 4:1

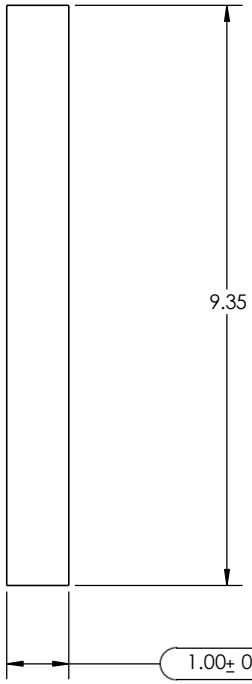


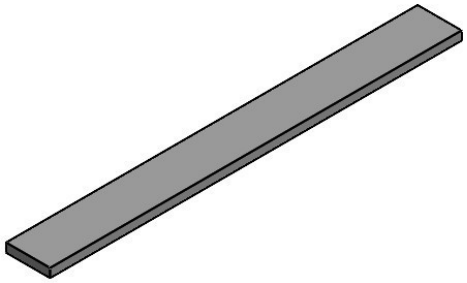
TOLERANCES UNLESS NOTED: UNITS:INCHES	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005 X.X ° °1"	UNIVERSITY OF COLORADO BOULDER
MATERIAL: STEEL		DESCRIPTION: HINGE ROD
FINISH: N/A	PN: LS04-01	REV: A
		SHEET: 1 of 1

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REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	11/23/2024





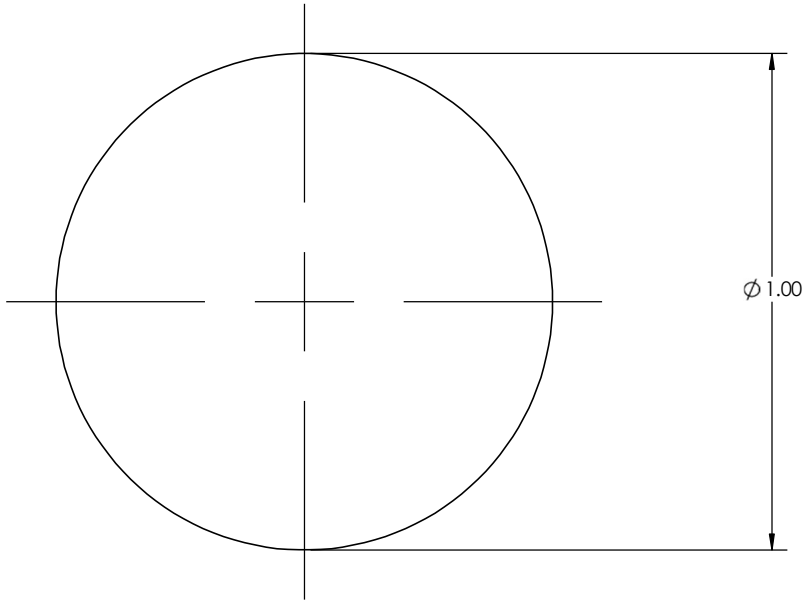
NOTE:
1.UNLESS METIONED SCALE IS 1:2

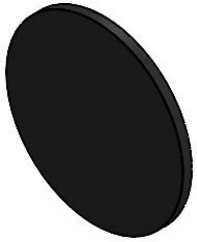
TOLERANCES UNLESS NOTED: UNITS:INCHES	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005 X.X ° ° 1"	UNIVERSITY OF COLORADO BOULDER	
MATERIAL: RUBBER		DESCRIPTION: Rubber Grip	
FINISH: N/A		PN: LS04-02	REV: A
			SHEET: 1 of 1

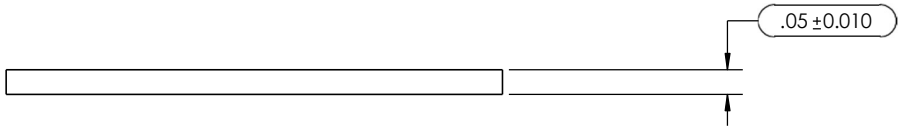
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REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	10/22/2024





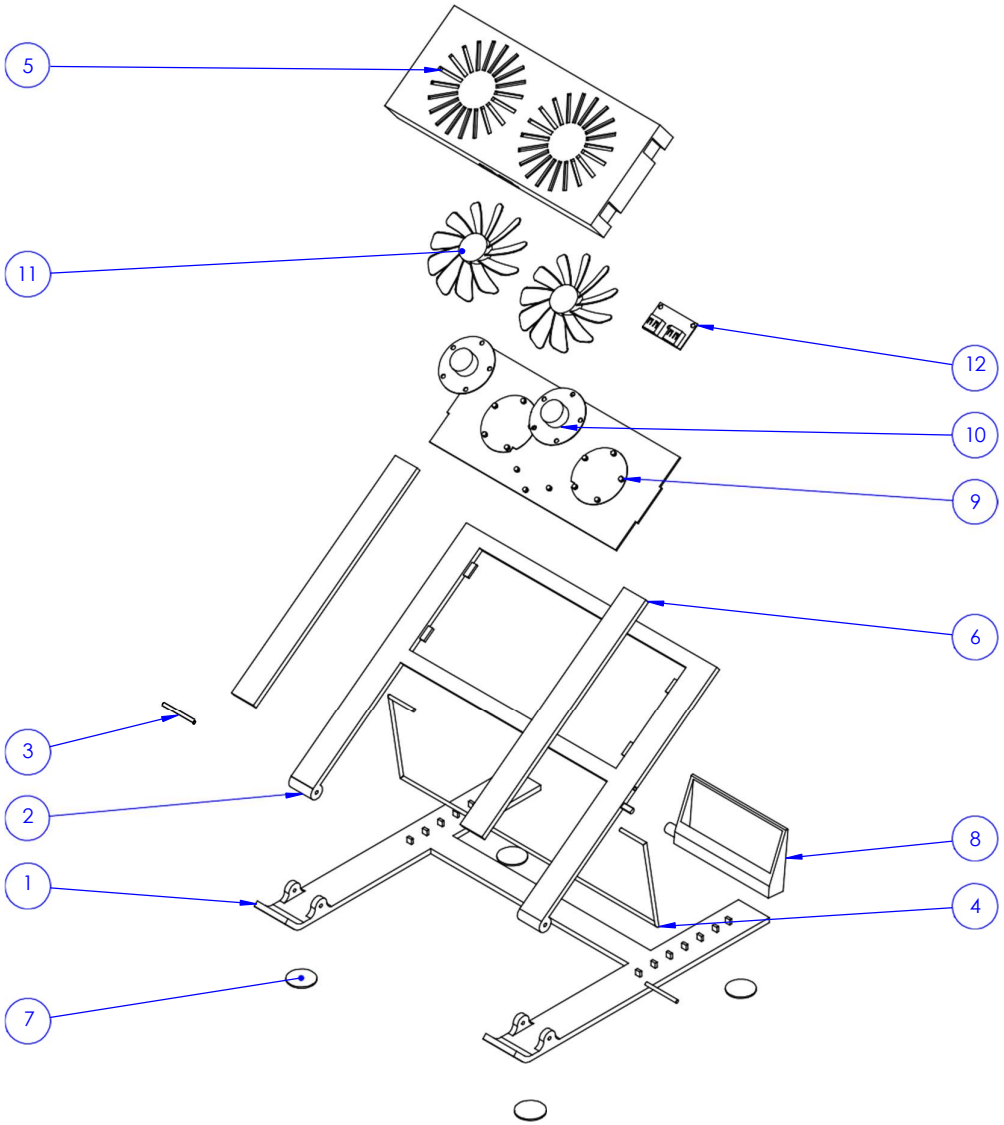


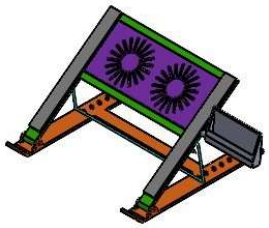
TOLERANCES UNLESS NOTED: UNITS:INCHES	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005 X.X ° °1"	UNIVERSITY OF COLORADO BOULDER	
MATERIAL: RUBBER		DESCRIPTION: Rubber grip Base	
FINISH: N/A		PN: LS04-04	REV: A
			SHEET: 1 of 1

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REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	11/25/2024





ITEM NO.	PART NUMB ER	Description	QTY.
1	LS02-01	Base Stand	1
2	LS02-02	Rise Stand	1
3	LS04-01	Hinge rod	2
4	LS02-03	Angle Support	1
5	LS01-02	Fan Housing-Top	1
6	LS04-02	Rubber grip	2
7	LS04-04	Rubber grip Base	4
8	LS03	Mobile Stand	1
9	LS01-03	Fan Housing-Bottom	1
10	LS01-04	Customized Motor-Controller	2
11	LS01-01	Fan	2
12	LS05	USB Module	1

UNIVERSITY O COLORADO BOULDER

DESCRIPTION
LAPTOP STAND WITH COOLING PAD

PN	REV	SHEET
LS-1	A	1 of 1

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